GRADE 3 COMMON CORE MATH

Answer Key

Main • Backward Branches • Forward 1 (Grade 4) • Forward 2 (Grade 5)

69 question sets • 1725 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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SET 1 • 3.OA.A.1 • OPERATIONS & ALGEBRAIC THINKING

Interpret a product as the total number of objects in equal groups

MAIN	Grade 3 main question	**3.OA.A.1**
BACKWARD 1	Find the total of an array by adding equal addends	**2.OA.C.4**
BACKWARD 2	Skip-count by 5s, 10s, and 100s	**2.NBT.A.2**
FORWARD 1	Interpret a multiplication equation as a comparison	**4.OA.A.1**
FORWARD 2	Interpret a fraction times a fractional quantity as equal parts of t…	**5.NF.B.4.a**

SET 1 • 3.OA.A.1

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

Which describes 4 × 6?

A. 4 groups with 6 objects in each group

B. 4 groups with 4 objects in each group

C. 6 more than 4

D. 4 objects and 6 objects

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • MAIN • Q1/5

9

QUESTION ID

S1-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. 4 groups with 6 objects in each group

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • MAIN • Q1/5

There are 3 bags with 9 marbles in each bag.

Write a multiplication expression for the total number of marbles. Then find the total.

Expression: _____

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • MAIN • Q2/5

10

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

3 × 9; 27 marbles

GRADING NOTE 9 × 3 is also correct for the expression. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • MAIN • Q2/5

True or false?

7 × 2 can mean 7 groups with 2 objects in each group.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • MAIN • Q3/5

11

QUESTION ID

S1-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • MAIN • Q3/5

Describe a situation that 6 × 5 could show.

Situation: _____

_____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • MAIN • Q4/5

12

QUESTION ID

S1-M-Q4

TYPE

Short answer

CORRECT ANSWER

6 boxes with 5 crayons in each box, 30 crayons in all.

GRADING NOTE Any situation with 6 equal groups of 5 (or 5 equal groups of 6) is correct. A situation that adds 6 and 5 is not correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • MAIN • Q4/5

True or false?

8 × 3 means 8 + 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • MAIN • Q5/5

13

QUESTION ID

S1-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

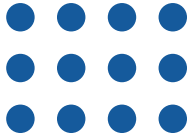
Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 1 • MAIN • Q5/5

SET 1 • 3.OA.A.1

BACKWARD 1

Find the total of an array by adding equal addends • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

Write an addition equation with equal addends for the array.



Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 1 • BACKWARD 1 • Q1/5

15

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 + 4 + 4 = 12

GRADING NOTE 3 + 3 + 3 + 3 = 12 (adding the columns) is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 1 • BACKWARD 1 • Q1/5

True or false?

An array with 2 rows of 5 dots has 10 dots.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 1 • BACKWARD 1 • Q2/5

16

QUESTION ID

S1-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 1 • BACKWARD 1 • Q2/5

Which addition equation matches an array with 4 rows of 3 dots?

A. 4 + 4 = 8

B. 4 + 3 = 7

C. 3 + 4 + 3 = 10

D. 3 + 3 + 3 + 3 = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 1 • BACKWARD 1 • Q3/5

17

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 + 3 + 3 + 3 = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 1 • BACKWARD 1 • Q3/5

How many dots are in an array with 5 rows of 5 dots?

Dots: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 1 • BACKWARD 1 • Q4/5

18

QUESTION ID

S1-B1-Q4

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 1 • BACKWARD 1 • Q4/5

True or false?

An array with 3 rows of 2 dots has 5 dots.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 1 • BACKWARD 1 • Q5/5

19

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 3.OA.A.1

BACKWARD 2

Skip-count by 5s, 10s, and 100s • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2

Skip-count by 5s.

5, 10, 15, ___, ___

Next two: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 1 • BACKWARD 2 • Q1/5

21

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20, 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 1 • BACKWARD 2 • Q1/5

True or false?

Skip-counting by 10s from 10 gives 10, 20, 30, 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 1 • BACKWARD 2 • Q2/5

22

QUESTION ID

S1-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 1 • BACKWARD 2 • Q2/5

Skip-count by 5s.

What number comes after 35?

A. 30

B. 36

C. 45

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 1 • BACKWARD 2 • Q3/5

23

QUESTION ID

S1-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 1 • BACKWARD 2 • Q3/5

Skip-count by 100s.

100, 200, 300, ___

Next: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 1 • BACKWARD 2 • Q4/5

24

QUESTION ID

S1-B2-Q4

TYPE

Short answer

CORRECT ANSWER

400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 1 • BACKWARD 2 • Q4/5

True or false?

Skip-counting by 5s from 5 gives 5, 10, 20, 25.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 1 • BACKWARD 2 • Q5/5

25

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 3.OA.A.1

FORWARD 1

Interpret a multiplication equation as a comparison • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1

Which statement matches 32 = 4 × 8?

A. 32 is 4 times as many as 8.

B. 32 is 4 more than 8.

C. 8 is 4 times as many as 32.

D. 32 is 8 more than 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • FORWARD 1 • Q1/5

27

QUESTION ID

S1-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. 32 is 4 times as many as 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • FORWARD 1 • Q1/5

Write a comparison statement for 42 = 6 × 7.

Statement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • FORWARD 1 • Q2/5

28

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

42 is 6 times as many as 7.

GRADING NOTE "42 is 7 times as many as 6" is also correct. A statement with "more than" is not correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • FORWARD 1 • Q2/5

True or false?

27 = 3 × 9 can mean that 27 is 3 times as many as 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • FORWARD 1 • Q3/5

29

QUESTION ID

S1-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation for "40 is 5 times as many as 8."

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • FORWARD 1 • Q4/5

30

QUESTION ID

S1-F1-Q4

TYPE

Short answer

CORRECT ANSWER

40 = 5 × 8

GRADING NOTE 40 = 8 × 5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • FORWARD 1 • Q4/5

True or false?

15 = 3 × 5 means that 15 is 3 more than 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • FORWARD 1 • Q5/5

31

QUESTION ID

S1-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • 3.OA.A.1

FORWARD 2

Interpret a fraction times a fractional quantity as equal parts of that quantity • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Which describes $\frac{2}{3} \times \frac{3}{4}$?

A. Split $\frac{3}{4}$ into 3 equal parts and take 2 of them

B. Split $\frac{3}{4}$ into 2 equal parts and take 3 of them

C. Add $\frac{2}{3}$ and $\frac{3}{4}$

D. Subtract $\frac{2}{3}$ from $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 1 • FORWARD 2 • Q1/5

33

QUESTION ID

S1-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Split 3/4 into 3 equal parts and take 2 of them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 1 • FORWARD 2 • Q1/5

Find $\frac{1}{2}$ of $\frac{4}{5}$ by splitting $\frac{4}{5}$ into 2 equal parts.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 1 • FORWARD 2 • Q2/5

34

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 1 • FORWARD 2 • Q2/5

True or false?

$\frac{1}{3} \times \frac{3}{8}$ is the size of one part when $\frac{3}{8}$ is split into 3 equal parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 1 • FORWARD 2 • Q3/5

35

QUESTION ID

S1-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 1 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{3}{4} \times \frac{2}{3}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 1 • FORWARD 2 • Q4/5

36

QUESTION ID

S1-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE 6/12 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 1 • FORWARD 2 • Q4/5

True or false?

$\frac{2}{5} \times \frac{5}{6} = \frac{10}{11}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 1 • FORWARD 2 • Q5/5

37

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 1 • FORWARD 2 • Q5/5

SET 2 • 3.OA.A.1 • OPERATIONS & ALGEBRAIC THINKING

Write a multiplication equation for equal groups and arrays

MAIN	Grade 3 main question	**3.OA.A.1**
BACKWARD 1	Write an addition equation with equal addends for an array	**2.OA.C.4**
BACKWARD 2	Add up to four two-digit numbers	**2.NBT.B.6**
FORWARD 1	Write an equation for a multiplicative comparison word problem	**4.OA.A.2**
FORWARD 2	Write expressions that record calculations with numbers	**5.OA.A.2**

SET 2 • 3.OA.A.1

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

Write a multiplication equation for the array.



Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • MAIN • Q1/5

40

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

3 × 7 = 21

GRADING NOTE 7 × 3 = 21 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • MAIN • Q1/5

A garden has 5 rows with 8 plants in each row.

Which equation gives the number of plants?

A. $5 \times 8 = 40$

B. $8 - 5 = 3$

C. $5 + 8 = 13$

D. $5 \times 5 = 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • MAIN • Q2/5

41

QUESTION ID

S2-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 5 × 8 = 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • MAIN • Q2/5

True or false?

The equation 6 × 4 = 24 matches 6 vases with 4 flowers in each vase.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • MAIN • Q3/5

42

QUESTION ID

S2-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • MAIN • Q3/5

Each of 7 spiders has 8 legs.

Write a multiplication equation for the total number of legs.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • MAIN • Q4/5

43

QUESTION ID

S2-M-Q4

TYPE

Short answer

CORRECT ANSWER

7 × 8 = 56

GRADING NOTE 8 × 7 = 56 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • MAIN • Q4/5

True or false?

The equation 4 × 9 = 36 matches 4 boxes with 4 pens in each box.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • MAIN • Q5/5

44

QUESTION ID

S2-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • MAIN • Q5/5

SET 2 • 3.OA.A.1

BACKWARD 1

Write an addition equation with equal addends for an array • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

Write an addition equation with equal addends for the array.



Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 2 • BACKWARD 1 • Q1/5

46

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5 + 5 = 10

GRADING NOTE 2 + 2 + 2 + 2 + 2 = 10 (adding the columns) is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 2 • BACKWARD 1 • Q1/5

True or false?

3 + 3 + 3 + 3 matches an array with 4 rows of 3 dots.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 2 • BACKWARD 1 • Q2/5

47

QUESTION ID

S2-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 2 • BACKWARD 1 • Q2/5

An array has 3 rows with 5 dots in each row.

Which equation gives the total?

A. 5 + 5 + 5 = 15

B. 5 + 5 = 10

C. 3 + 3 = 6

D. 3 + 5 = 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 2 • BACKWARD 1 • Q3/5

48

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5 + 5 + 5 = 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 2 • BACKWARD 1 • Q3/5

An array has 4 rows with 4 dots in each row.

Write an addition equation for the total.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 2 • BACKWARD 1 • Q4/5

49

QUESTION ID

S2-B1-Q4

TYPE

Short answer

CORRECT ANSWER

4 + 4 + 4 + 4 = 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 2 • BACKWARD 1 • Q4/5

True or false?

An array with 5 rows of 2 dots matches 5 + 2 = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 2 • BACKWARD 1 • Q5/5

50

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 3.OA.A.1

BACKWARD 2

Add up to four two-digit numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6

Add.

15 + 15 + 15

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 2 • BACKWARD 2 • Q1/5

52

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 2 • BACKWARD 2 • Q1/5

True or false?

20 + 20 + 20 + 20 = 80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 2 • BACKWARD 2 • Q2/5

53

QUESTION ID

S2-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 2 • BACKWARD 2 • Q2/5

11 + 11 + 11 + 11 = ?

A. 22

B. 44

C. 40

D. 33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 2 • BACKWARD 2 • Q3/5

54

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 44

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 2 • BACKWARD 2 • Q3/5

Add.

24 + 24

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 2 • BACKWARD 2 • Q4/5

55

QUESTION ID

S2-B2-Q4

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 2 • BACKWARD 2 • Q4/5

True or false?

13 + 13 + 13 = 36

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 2 • BACKWARD 2 • Q5/5

56

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 3.OA.A.1

FORWARD 1

Write an equation for a multiplicative comparison word problem • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A cat weighs 7 pounds. A dog weighs 5 times as much.

Write an equation with a symbol for the dog's weight. Then solve it.

Equation: _____

Weight: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 2 • FORWARD 1 • Q1/5

58

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

d = 5 × 7; 35 pounds

GRADING NOTE Any equivalent equation with a symbol for the unknown is correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 2 • FORWARD 1 • Q1/5

True or false?

Ann has 6 shells. Ben has 3 times as many shells. Ben has 18 shells.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 2 • FORWARD 1 • Q2/5

59

QUESTION ID

S2-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 2 • FORWARD 1 • Q2/5

A red tower is 9 blocks tall. A blue tower is 4 times as tall.

Which equation gives the blue tower's height, b?

A. b = 9 ÷ 4

B. b = 4 + 9

C. b = 4 × 9

D. b = 9 − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 2 • FORWARD 1 • Q3/5

60

QUESTION ID

S2-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. b = 4 × 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 2 • FORWARD 1 • Q3/5

A plant is 8 cm tall. A tree is 10 times as tall.

How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 2 • FORWARD 1 • Q4/5

61

QUESTION ID

S2-F1-Q4

TYPE

Short answer

CORRECT ANSWER

80 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 2 • FORWARD 1 • Q4/5

True or false?

A rope is 6 feet long. A rope 3 times as long is 9 feet long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 2 • FORWARD 1 • Q5/5

62

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 3.OA.A.1

FORWARD 2

Write expressions that record calculations with numbers • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for "multiply 4 by the sum of 6 and 3."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 2 • Q1/5

64

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 × (6 + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 2 • Q1/5

Which expression means "3 times the difference of 10 and 2"?

A. $(3 \times 10) - 2$

B. $10 - 2 \times 3$

C. $3 \times (10 - 2)$

D. $3 \times 10 - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 2 • Q2/5

65

QUESTION ID

S2-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $3 \times (10 - 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 2 • Q2/5

True or false?

5 × (7 + 2) is 5 times as large as 7 + 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 2 • Q3/5

66

QUESTION ID

S2-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 2 • Q3/5

There are 4 boxes. Each box has 6 red pens and 2 blue pens.

Write an expression for the total number of pens.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 2 • Q4/5

67

QUESTION ID

S2-F2-Q4

TYPE

Short answer

CORRECT ANSWER

4 × (6 + 2)

GRADING NOTE (6 + 2) × 4 and 4 × 6 + 4 × 2 are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 2 • Q4/5

True or false?

2 × (8 + 1) is 2 more than 8 + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 2 • Q5/5

68

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 2 • FORWARD 2 • Q5/5

SET 3 • 3.OA.A.2 • OPERATIONS & ALGEBRAIC THINKING

Interpret a quotient as the number of objects in each share

MAIN	Grade 3 main question	**3.OA.A.2**
BACKWARD 1	Split a group of up to 20 objects into two equal groups	**2.OA.C.3**
BACKWARD 2	Split a rectangle into rows and columns of same-size squares and…	**2.G.A.2**
FORWARD 1	Solve multiplicative comparison word problems by dividing	**4.OA.A.2**
FORWARD 2	Interpret a fraction as division to find equal shares	**5.NF.B.3**

SET 3 • 3.OA.A.2

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

28 stickers are shared equally among 4 friends.

What does 28 ÷ 4 tell you?

A. The number of friends

B. The number of stickers each friend gets

C. The total number of stickers

D. The number of stickers left over

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • MAIN • Q1/5

71

QUESTION ID

S3-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. The number of stickers each friend gets

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • MAIN • Q1/5

18 apples are shared equally among 3 baskets.

Write a division equation for the number of apples in each basket. Then solve it.

Equation: _____

Apples in each basket: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • MAIN • Q2/5

72

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

18 ÷ 3 = 6; 6 apples

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • MAIN • Q2/5

True or false?

In 35 ÷ 5 = 7, the 7 can mean the number in each group when 35 objects are shared equally among 5 groups.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • MAIN • Q3/5

73

QUESTION ID

S3-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe a situation in which 24 ÷ 6 tells how many objects are in each share.

Situation: _____

_____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • MAIN • Q4/5

74

QUESTION ID

S3-M-Q4

TYPE

Short answer

CORRECT ANSWER

24 cookies are shared equally by 6 children, so each child gets 4 cookies.

GRADING NOTE Any situation in which 24 objects are shared equally among 6 groups, with the answer as the number in each group, is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • MAIN • Q4/5

True or false?

When 20 pencils are shared equally among 4 students, each student gets 16 pencils.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • MAIN • Q5/5

75

QUESTION ID

S3-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • MAIN • Q5/5

SET 3 • 3.OA.A.2

BACKWARD 1

Split a group of up to 20 objects into two equal groups • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

Write 16 as the sum of two equal addends.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 3 • BACKWARD 1 • Q1/5

77

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8 + 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 3 • BACKWARD 1 • Q1/5

True or false?

14 objects can be split into 2 equal groups of 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 3 • BACKWARD 1 • Q2/5

78

QUESTION ID

S3-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 3 • BACKWARD 1 • Q2/5

Which number of objects can be split into 2 equal groups?

A. 12

B. 9

C. 7

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 3 • BACKWARD 1 • Q3/5

79

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 3 • BACKWARD 1 • Q3/5

12 counters are split into 2 equal groups.

How many counters are in each group?

Counters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 3 • BACKWARD 1 • Q4/5

80

QUESTION ID

S3-B1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 3 • BACKWARD 1 • Q4/5

True or false?

11 objects can be split into 2 equal groups with none left over.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 3 • BACKWARD 1 • Q5/5

81

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 3.OA.A.2

BACKWARD 2

Split a rectangle into rows and columns of same-size squares and count them • Grade 2 • Geometry • 2.G.A.2

A rectangle is split into 3 rows of 4 same-size squares.

How many squares are there?

Squares: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 2 • Geometry • 2.G.A.2 • Set 3 • BACKWARD 2 • Q1/5

83

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 2 • Geometry • 2.G.A.2 • Set 3 • BACKWARD 2 • Q1/5

True or false?

A rectangle split into 2 rows of 5 same-size squares has 10 squares.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 2 • Geometry • 2.G.A.2 • Set 3 • BACKWARD 2 • Q2/5

84

QUESTION ID

S3-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 2 • Geometry • 2.G.A.2 • Set 3 • BACKWARD 2 • Q2/5

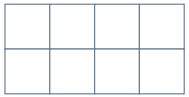
How many same-size squares are in the rectangle?

A. 8

B. 6

C. 10

D. 4



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 2 • Geometry • 2.G.A.2 • Set 3 • BACKWARD 2 • Q3/5

85

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

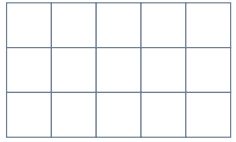
A. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 2 • Geometry • 2.G.A.2 • Set 3 • BACKWARD 2 • Q3/5

How many same-size squares are in each row of the rectangle?



Squares in each row: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 2 • Geometry • 2.G.A.2 • Set 3 • BACKWARD 2 • Q4/5

86

QUESTION ID

S3-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 2 • Geometry • 2.G.A.2 • Set 3 • BACKWARD 2 • Q4/5

True or false?

A rectangle split into 4 rows of 4 same-size squares has 8 squares.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 2 • Geometry • 2.G.A.2 • Set 3 • BACKWARD 2 • Q5/5

87

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 2 • Geometry • 2.G.A.2 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • 3.OA.A.2

FORWARD 1

Solve multiplicative comparison word problems by dividing • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A tree is 24 feet tall. That is 4 times as tall as a bush.

How tall is the bush?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • FORWARD 1 • Q1/5

89

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • FORWARD 1 • Q1/5

True or false?

Mia has 45 beads. That is 5 times as many beads as Jo has. Jo has 9 beads.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • FORWARD 1 • Q2/5

90

QUESTION ID

S3-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • FORWARD 1 • Q2/5

A truck weighs 6 tons. That is 3 times as much as a van weighs.

How much does the van weigh?

A. 18 tons

B. 2 tons

C. 3 tons

D. 9 tons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • FORWARD 1 • Q3/5

91

QUESTION ID

S3-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2 tons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • FORWARD 1 • Q3/5

A book costs $36. That is 9 times as much as a pen costs.

How much does the pen cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • FORWARD 1 • Q4/5

92

QUESTION ID

S3-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • FORWARD 1 • Q4/5

True or false?

A barn is 40 feet tall. That is 8 times as tall as a fence. The fence is 32 feet tall.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • FORWARD 1 • Q5/5

93

QUESTION ID

S3-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • 3.OA.A.2

FORWARD 2

Interpret a fraction as division to find equal shares • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

4 friends share 3 pizzas equally.

How much pizza does each friend get?

Pizza: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 3 • FORWARD 2 • Q1/5

95

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3/4 of a pizza

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 3 • FORWARD 2 • Q1/5

True or false?

When 2 sandwiches are shared equally by 5 people, each person gets $\frac{2}{5}$ of a sandwich.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 3 • FORWARD 2 • Q2/5

96

QUESTION ID

S3-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 3 • FORWARD 2 • Q2/5

6 people share 5 pounds of rice equally.

How much rice does each person get?

A. $\frac{5}{6}$ pound

B. $\frac{6}{5}$ pounds

C. 1 pound

D. $\frac{1}{6}$ pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 3 • FORWARD 2 • Q3/5

97

QUESTION ID

S3-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5/6 pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 3 • FORWARD 2 • Q3/5

Write 7 ÷ 3 as a fraction or a mixed number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 3 • FORWARD 2 • Q4/5

98

QUESTION ID

S3-F2-Q4

TYPE

Short answer

CORRECT ANSWER

7/3

GRADING NOTE 2 1/3 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 3 • FORWARD 2 • Q4/5

True or false?

$3 \div 8 = \frac{8}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 3 • FORWARD 2 • Q5/5

99

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.3 • Set 3 • FORWARD 2 • Q5/5

SET 4 • 3.OA.A.2 • OPERATIONS & ALGEBRAIC THINKING

Interpret a quotient as the number of equal shares

MAIN	Grade 3 main question	**3.OA.A.2**
BACKWARD 1	Skip-count by 5s and 10s	**2.NBT.A.2**
BACKWARD 2	Rows and columns of arrays up to 5 by 5	**2.OA.C.4**
FORWARD 1	Solve multistep word problems that require interpreting a remaind…	**4.OA.A.3**
FORWARD 2	Divide a whole number by a unit fraction to find the number of eq…	**5.NF.B.7.c**

SET 4 • 3.OA.A.2

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

30 pencils are put in boxes, with 5 pencils in each box.

What does 30 ÷ 5 tell you?

A. The number of pencils left over

B. The number of boxes

C. The number of pencils in each box

D. The total number of pencils

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • MAIN • Q1/5

102

QUESTION ID

S4-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. The number of boxes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • MAIN • Q1/5

24 eggs are packed 6 to a carton.

Write a division equation for the number of cartons.
Then solve it.

Equation: _____

Cartons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • MAIN • Q2/5

103

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

24 ÷ 6 = 4; 4 cartons

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • MAIN • Q2/5

True or false?

In 40 ÷ 8 = 5, the 5 can mean the number of groups when 40 objects are put into groups of 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • MAIN • Q3/5

104

QUESTION ID

S4-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • MAIN • Q3/5

Describe a situation in which 21 ÷ 3 tells how many groups there are.

Situation: _____

_____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • MAIN • Q4/5

105

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

21 students make teams of 3, so there are 7 teams.

GRADING NOTE Any situation in which 21 objects are put into groups of 3, with the answer as the number of groups, is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • MAIN • Q4/5

True or false?

A coach puts 36 balls into bags of 4. There are 32 bags.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • MAIN • Q5/5

106

QUESTION ID

S4-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • MAIN • Q5/5

SET 4 • 3.OA.A.2

BACKWARD 1

Skip-count by 5s and 10s • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2

Skip-count by 10s from 0 to 60.

How many 10s did you count?

Tens: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 4 • BACKWARD 1 • Q1/5

108

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 4 • BACKWARD 1 • Q1/5

True or false?

Counting by 5s from 0 to 25 takes 5 counts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 4 • BACKWARD 1 • Q2/5

109

QUESTION ID

S4-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 4 • BACKWARD 1 • Q2/5

Starting at 0, how many 5s do you count to reach 40?

A. 35

B. 8

C. 40

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 4 • BACKWARD 1 • Q3/5

110

QUESTION ID

S4-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 4 • BACKWARD 1 • Q3/5

Skip-count by 10s.

40, 50, 60, ___, ___

Next two: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 4 • BACKWARD 1 • Q4/5

111

QUESTION ID

S4-B1-Q4

TYPE

Short answer

CORRECT ANSWER

70, 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 4 • BACKWARD 1 • Q4/5

True or false?

Counting by 10s from 0 to 90 takes 10 counts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 4 • BACKWARD 1 • Q5/5

112

QUESTION ID

S4-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

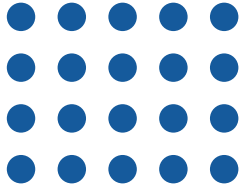
Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 3.OA.A.2

BACKWARD 2

Rows and columns of arrays up to 5 by 5 • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

How many rows does the array have? How many dots are in each row?



Rows: _____

Dots in each row: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 4 • BACKWARD 2 • Q1/5

114

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4 rows; 5 dots

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 4 • BACKWARD 2 • Q1/5

True or false?

The array has 3 rows.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 4 • BACKWARD 2 • Q2/5

115

QUESTION ID

S4-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 4 • BACKWARD 2 • Q2/5

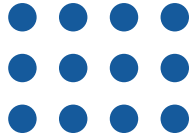
How many dots are in each row of the array?

A. 12

B. 4

C. 3

D. 7



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 4 • BACKWARD 2 • Q3/5

116

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 4 • BACKWARD 2 • Q3/5

Write an addition equation with equal addends for the array.



Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 4 • BACKWARD 2 • Q4/5

117

QUESTION ID

S4-B2-Q4

TYPE

Short answer

CORRECT ANSWER

4 + 4 = 8

GRADING NOTE 2 + 2 + 2 + 2 = 8 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 4 • BACKWARD 2 • Q4/5

True or false?

The array has 5 rows.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 4 • BACKWARD 2 • Q5/5

118

QUESTION ID

S4-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 3.OA.A.2

FORWARD 1

Solve multistep word problems that require interpreting a remainder • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

45 students are going on a trip, but 2 are absent. The rest ride in vans that hold 6 students each.

How many vans are needed?

Vans: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 4 • FORWARD 1 • Q1/5

120

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8 vans

GRADING NOTE 45 − 2 = 43 and 43 ÷ 6 = 7 R 1. The 1 extra student needs another van, so 8 vans (not 7, and not 7 R 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 4 • FORWARD 1 • Q1/5

A baker makes 3 trays of 18 cookies. She packs them in bags of 8.

How many full bags can she make? How many cookies are left over?

Full bags: _____

Left over: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 4 • FORWARD 1 • Q2/5

121

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6 full bags; 6 cookies

GRADING NOTE 3 × 18 = 54 and 54 ÷ 8 = 6 R 6. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 4 • FORWARD 1 • Q2/5

A club has $90. It spends $20 on snacks and the rest on tickets that cost $8 each.

How many tickets can it buy?

A. 8

B. 9

C. 11

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 4 • FORWARD 1 • Q3/5

122

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 4 • FORWARD 1 • Q3/5

A farmer has 4 crates of 25 apples. He puts all the apples in boxes that hold 6 apples each.

Write an equation with a letter for the number of apples. Then find the fewest boxes he needs.

Equation: _____

Boxes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 4 • FORWARD 1 • Q4/5

123

QUESTION ID

S4-F1-Q4

TYPE

Short answer

CORRECT ANSWER

a = 4 × 25, so

a = 100; 17 boxes

GRADING NOTE Any equivalent equation with a letter is correct. 100 ÷ 6 = 16 R 4, and the 4 extra apples need another box. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 4 • FORWARD 1 • Q4/5

True or false?

Ana has 60 feet of ribbon. She uses 7 feet and cuts the rest into 6-foot pieces. She gets 8 full pieces, and 5 feet are left over.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 4 • FORWARD 1 • Q5/5

124

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 4 • FORWARD 1 • Q5/5

SET 4 • 3.OA.A.2

FORWARD 2

Divide a whole number by a unit fraction to find the number of equal shares • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c

How many $\frac{1}{3}$-cup servings are in 4 cups?

Servings: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 4 • FORWARD 2 • Q1/5

126

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 4 • FORWARD 2 • Q1/5

True or false?

A 6-foot board is cut into pieces that are each $\frac{1}{2}$ foot long. There are 12 pieces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 4 • FORWARD 2 • Q2/5

127

QUESTION ID

S4-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 4 • FORWARD 2 • Q2/5

A 5-foot board is cut into pieces that are each $\frac{1}{4}$ foot long.

How many pieces are there?

A. $\frac{1}{20}$

B. $\frac{5}{4}$

C. 20

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 4 • FORWARD 2 • Q3/5

128

QUESTION ID

S4-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 4 • FORWARD 2 • Q3/5

How many $\frac{1}{5}$-pound bags can be filled from 3 pounds of trail mix?

Bags: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 4 • FORWARD 2 • Q4/5

129

QUESTION ID

S4-F2-Q4

TYPE

Short answer

CORRECT ANSWER

15 bags

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 4 • FORWARD 2 • Q4/5

True or false?

2 liters of juice exactly fill 6 glasses that each hold $\frac{1}{4}$ liter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 4 • FORWARD 2 • Q5/5

130

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c • Set 4 • FORWARD 2 • Q5/5

SET 5 • 3.OA.A.3 • OPERATIONS & ALGEBRAIC THINKING

Solve multiplication word problems within 100

MAIN	Grade 3 main question	**3.OA.A.3**
BACKWARD 1	One-step addition word problems within 100	**2.OA.A.1**
BACKWARD 2	Add lengths in word problems within 100	**2.MD.B.5**
FORWARD 1	Multiply a two-digit number by a one-digit number to solve word…	**4.NBT.B.5**
FORWARD 2	Multiply multi-digit whole numbers to solve word problems	**5.NBT.B.5**

SET 5 • 3.OA.A.3

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

A bakery puts 6 muffins in each box. It fills 7 boxes.

How many muffins are in the boxes?

Muffins: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 5 • MAIN • Q1/5

133

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

42 muffins

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 5 • MAIN • Q1/5

A wall has 8 rows of tiles with 9 tiles in each row.

How many tiles are on the wall?

Tiles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 5 • MAIN • Q2/5

134

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

72 tiles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 5 • MAIN • Q2/5

Each ribbon is 4 feet long. Mia has 6 ribbons.

What is the total length of the ribbons?

A. 28 feet

B. 24 feet

C. 2 feet

D. 10 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 5 • MAIN • Q3/5

135

QUESTION ID

S5-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 24 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 5 • MAIN • Q3/5

True or false?

A parking lot has 5 rows with 9 cars in each row. There are 54 cars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 5 • MAIN • Q4/5

136

QUESTION ID

S5-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 5 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Tickets cost $7 each. A family buys 8 tickets.

Write an equation with a symbol for the total cost. Then solve it.

Equation: _____

Total cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 5 • MAIN • Q5/5

137

QUESTION ID

S5-M-Q5

TYPE

Short answer

CORRECT ANSWER

8 × 7 = c; $56

GRADING NOTE Any equation with a symbol for the unknown (such as 8 × 7 = ?) is correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 5 • MAIN • Q5/5

SET 5 • 3.OA.A.3

BACKWARD 1

One-step addition word problems within 100 • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

A box has 24 crayons. Another box has 24 crayons.

How many crayons are there in all?

Crayons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q1/5

139

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q1/5

True or false?

Jo has 35 cards and gets 15 more. She has 50 cards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q2/5

140

QUESTION ID

S5-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q2/5

A bus has 27 children and 18 adults.

How many people are on the bus?

A. 45

B. 35

C. 9

D. 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q3/5

141

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q3/5

A farmer picks 46 apples in the morning and 38 apples in the afternoon.

How many apples does he pick?

Apples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q4/5

142

QUESTION ID

S5-B1-Q4

TYPE

Short answer

CORRECT ANSWER

84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q4/5

True or false?

A class has 16 girls and 17 boys. There are 23 students.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q5/5

143

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 3.OA.A.3

BACKWARD 2

Add lengths in word problems within 100 • Grade 2 • Measurement & Data • 2.MD.B.5

A red ribbon is 25 cm long. A blue ribbon is 30 cm long.

What is their total length?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.5 • Set 5 • BACKWARD 2 • Q1/5

145

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

55 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.5 • Set 5 • BACKWARD 2 • Q1/5

True or false?

Two 20-foot ropes laid end to end are 40 feet long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.5 • Set 5 • BACKWARD 2 • Q2/5

146

QUESTION ID

S5-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.5 • Set 5 • BACKWARD 2 • Q2/5

One path is 48 m long and another is 36 m long.

What is their total length?

A. 74 m

B. 12 m

C. 94 m

D. 84 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.5 • Set 5 • BACKWARD 2 • Q3/5

147

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 84 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.5 • Set 5 • BACKWARD 2 • Q3/5

A board is 47 inches long. Ben glues a 25-inch board to its end.

How long is it now?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.5 • Set 5 • BACKWARD 2 • Q4/5

148

QUESTION ID

S5-B2-Q4

TYPE

Short answer

CORRECT ANSWER

72 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.5 • Set 5 • BACKWARD 2 • Q4/5

True or false?

A 60-cm stick and a 35-cm stick laid end to end are 85 cm long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.5 • Set 5 • BACKWARD 2 • Q5/5

149

QUESTION ID

S5-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.5 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 3.OA.A.3

FORWARD 1

Multiply a two-digit number by a one-digit number to solve word problems • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

A school buys 6 boxes of 24 pencils.

How many pencils does it buy?

Pencils: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 5 • FORWARD 1 • Q1/5

151

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

144 pencils

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 5 • FORWARD 1 • Q1/5

True or false?

A theater has 9 rows of 32 seats. It has 288 seats.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 5 • FORWARD 1 • Q2/5

152

QUESTION ID

S5-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 5 • FORWARD 1 • Q2/5

A bike trip covers 45 miles each day for 4 days.

How long is the trip?

A. 180 miles

B. 49 miles

C. 90 miles

D. 160 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 5 • FORWARD 1 • Q3/5

153

QUESTION ID

S5-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 180 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 5 • FORWARD 1 • Q3/5

Tickets cost $18 each. A club buys 7 tickets.

What is the total cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 5 • FORWARD 1 • Q4/5

154

QUESTION ID

S5-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$126

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 5 • FORWARD 1 • Q4/5

True or false?

A farmer plants 5 rows of 76 trees. He plants 350 trees.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 5 • FORWARD 1 • Q5/5

155

QUESTION ID

S5-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 5 • FORWARD 1 • Q5/5

SET 5 • 3.OA.A.3

FORWARD 2

Multiply multi-digit whole numbers to solve word problems • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5

A stadium has 48 sections with 125 seats in each section.

How many seats are there?

Seats: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 5 • FORWARD 2 • Q1/5

157

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

6,000 seats

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 5 • FORWARD 2 • Q1/5

True or false?

A truck carries 36 crates with 144 apples in each crate. It carries 5,184 apples.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 5 • FORWARD 2 • Q2/5

158

QUESTION ID

S5-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 5 • FORWARD 2 • Q2/5

A printer prints 215 pages each hour for 24 hours.

How many pages does it print?

A. 4,300

B. 5,060

C. 5,160

D. 1,290

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 5 • FORWARD 2 • Q3/5

159

QUESTION ID

S5-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5,160

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 5 • FORWARD 2 • Q3/5

A school orders 32 boxes with 250 sheets of paper in each box.

How many sheets is that?

Sheets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 5 • FORWARD 2 • Q4/5

160

QUESTION ID

S5-F2-Q4

TYPE

Short answer

CORRECT ANSWER

8,000 sheets

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 5 • FORWARD 2 • Q4/5

True or false?

A farm packs 64 cartons of 18 eggs. That is 1,052 eggs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 5 • FORWARD 2 • Q5/5

161

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 5 • FORWARD 2 • Q5/5

SET 6 • 3.OA.A.3 • OPERATIONS & ALGEBRAIC THINKING

Solve division word problems within 100

MAIN	Grade 3 main question	**3.OA.A.3**
BACKWARD 1	One-step subtraction word problems within 100	**2.OA.A.1**
BACKWARD 2	Split circles and rectangles into 2, 3, or 4 equal shares	**2.G.A.3**
FORWARD 1	Divide up to three-digit numbers by one-digit numbers to solve w…	**4.NBT.B.6**
FORWARD 2	Divide by two-digit divisors to solve word problems	**5.NBT.B.6**

SET 6 • 3.OA.A.3

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

56 students are split equally into 7 teams.

How many students are on each team?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 6 • MAIN • Q1/5

164

QUESTION ID

S6-M-Q1

TYPE

Short answer

CORRECT ANSWER

8 students

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 6 • MAIN • Q1/5

A gardener plants 63 seeds in rows of 9.

How many rows does she plant?

Rows: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 6 • MAIN • Q2/5

165

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

7 rows

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 6 • MAIN • Q2/5

A 48-inch ribbon is cut into 6 equal pieces.

How long is each piece?

A. 42 inches

B. 8 inches

C. 6 inches

D. 54 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 6 • MAIN • Q3/5

166

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 6 • MAIN • Q3/5

True or false?

81 chairs are set up in 9 equal rows. There are 9 chairs in each row.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 6 • MAIN • Q4/5

167

QUESTION ID

S6-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 6 • MAIN • Q4/5

A coach has 54 balls. She puts the same number of balls in each of 6 bags.

Write an equation with a symbol for the number of balls in each bag. Then solve it.

Equation: _____

Balls in each bag: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 6 • MAIN • Q5/5

168

QUESTION ID

S6-M-Q5

TYPE

Short answer

CORRECT ANSWER

54 ÷ 6 = b; 9 balls

GRADING NOTE Any equation with a symbol for the unknown (such as 6 × b = 54) is correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 6 • MAIN • Q5/5

SET 6 • 3.OA.A.3

BACKWARD 1

One-step subtraction word problems within 100 • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

A shelf has 64 books. 28 books are checked out.

How many books are left?

Books: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 6 • BACKWARD 1 • Q1/5

170

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 6 • BACKWARD 1 • Q1/5

True or false?

Mo had 50 stickers and gave away 18. He has 32 stickers left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 6 • BACKWARD 1 • Q2/5

171

QUESTION ID

S6-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 6 • BACKWARD 1 • Q2/5

A tank has 90 fish. 45 fish are moved to another tank.

How many fish are left?

A. 35

B. 135

C. 55

D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 6 • BACKWARD 1 • Q3/5

172

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 6 • BACKWARD 1 • Q3/5

A rope is 75 feet long. 30 feet are cut off.

How much rope is left?

Rope left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 6 • BACKWARD 1 • Q4/5

173

QUESTION ID

S6-B1-Q4

TYPE

Short answer

CORRECT ANSWER

45 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 6 • BACKWARD 1 • Q4/5

True or false?

A box had 40 pears. 15 were eaten, so 35 are left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 6 • BACKWARD 1 • Q5/5

174

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 3.OA.A.3

BACKWARD 2

Split circles and rectangles into 2, 3, or 4 equal shares • Grade 2 • Geometry • 2.G.A.3

A rectangle is split into 4 equal shares.

What is each share called?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 2 • Geometry • 2.G.A.3 • Set 6 • BACKWARD 2 • Q1/5

176

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

A fourth

GRADING NOTE "A quarter" is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 2 • Geometry • 2.G.A.3 • Set 6 • BACKWARD 2 • Q1/5

True or false?

A circle cut into 3 equal shares is cut into thirds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 2 • Geometry • 2.G.A.3 • Set 6 • BACKWARD 2 • Q2/5

177

QUESTION ID

S6-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 2 • Geometry • 2.G.A.3 • Set 6 • BACKWARD 2 • Q2/5

A sandwich is cut into 2 equal shares.

What is each share called?

A. A third

B. A whole

C. A fourth

D. A half

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 2 • Geometry • 2.G.A.3 • Set 6 • BACKWARD 2 • Q3/5

178

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A half

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 2 • Geometry • 2.G.A.3 • Set 6 • BACKWARD 2 • Q3/5

How many thirds make a whole?

Thirds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 2 • Geometry • 2.G.A.3 • Set 6 • BACKWARD 2 • Q4/5

179

QUESTION ID

S6-B2-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q4

Grade 2 • Geometry • 2.G.A.3 • Set 6 • BACKWARD 2 • Q4/5

True or false?

A rectangle cut into 4 shares of different sizes is cut into fourths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 2 • Geometry • 2.G.A.3 • Set 6 • BACKWARD 2 • Q5/5

180

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 2 • Geometry • 2.G.A.3 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 3.OA.A.3

FORWARD 1

Divide up to three-digit numbers by one-digit numbers to solve word problems • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

A farmer packs 96 eggs into cartons of 6.

How many cartons does he fill?

Cartons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 6 • FORWARD 1 • Q1/5

182

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

16 cartons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 6 • FORWARD 1 • Q1/5

True or false?

A 135-page book is read in 5 equal parts. Each part is 27 pages.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 6 • FORWARD 1 • Q2/5

183

QUESTION ID

S6-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 6 • FORWARD 1 • Q2/5

A school splits 252 students into 4 equal groups.

How many students are in each group?

A. 248

B. 61

C. 63

D. 1,008

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 6 • FORWARD 1 • Q3/5

184

QUESTION ID

S6-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 63

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 6 • FORWARD 1 • Q3/5

A rope 128 feet long is cut into 8 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 6 • FORWARD 1 • Q4/5

185

QUESTION ID

S6-F1-Q4

TYPE

Short answer

CORRECT ANSWER

16 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 6 • FORWARD 1 • Q4/5

True or false?

7 friends share 196 marbles equally. Each friend gets 29 marbles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 6 • FORWARD 1 • Q5/5

186

QUESTION ID

S6-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 3.OA.A.3

FORWARD 2

Divide by two-digit divisors to solve word problems • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

A school has 672 students in 24 classes of the same size.

How many students are in each class?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 6 • FORWARD 2 • Q1/5

188

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 6 • FORWARD 2 • Q1/5

True or false?

A 1,350-mile trip is driven in 15 days, the same distance each day. Each day is 90 miles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 6 • FORWARD 2 • Q2/5

189

QUESTION ID

S6-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 6 • FORWARD 2 • Q2/5

A factory packs 2,016 cans in boxes of 48.

How many boxes does it fill?

A. 43

B. 42

C. 2,064

D. 41

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 6 • FORWARD 2 • Q3/5

190

QUESTION ID

S6-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 6 • FORWARD 2 • Q3/5

An orchard has 1,155 trees in 35 equal rows.

How many trees are in each row?

Trees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 6 • FORWARD 2 • Q4/5

191

QUESTION ID

S6-F2-Q4

TYPE

Short answer

CORRECT ANSWER

33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 6 • FORWARD 2 • Q4/5

True or false?

3,600 seats are split into 25 equal sections. Each section has 154 seats.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 6 • FORWARD 2 • Q5/5

192

QUESTION ID

S6-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 6 • FORWARD 2 • Q5/5

SET 7 • 3.OA.A.3 • OPERATIONS & ALGEBRAIC THINKING

Represent a multiplication or division word problem with a drawing and an equation

MAIN	Grade 3 main question	**3.OA.A.3**
BACKWARD 1	Use an equation with a symbol for the unknown in a one-step wo…	**2.OA.A.1**
BACKWARD 2	Draw arrays with up to 5 rows and 5 columns	**2.OA.C.4**
FORWARD 1	Use a drawing and an equation to solve a multiplicative comparis…	**4.OA.A.2**
FORWARD 2	Use a tape diagram to find a fraction of a fractional or mixed-nu…	**5.NF.B.6**

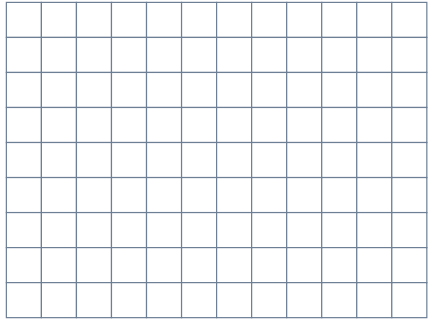
SET 7 • 3.OA.A.3

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

There are 4 plates with 5 grapes on each plate.

Draw a picture of the problem. Then write an equation and find the total.



Equation: _____

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 7 • MAIN • Q1/5

195

QUESTION ID

S7-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: 4 equal groups with 5 objects in each (or a 4-by-5 array, or another diagram of 4 equal groups of 5)

Written: 4 × 5 = 20; 20 grapes

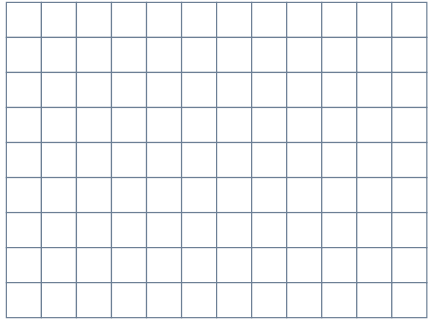
GRADING NOTE Grade both: a drawing that shows 4 equal groups of 5, and the equation with the total. 5 × 4 = 20 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 7 • MAIN • Q1/5

18 crackers are shared equally among 3 children.

Draw a picture of the problem. Then write a division equation and find how many crackers each child gets.



Equation: _____

Each child: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 7 • MAIN • Q2/5

196

QUESTION ID

S7-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: 18 objects split into 3 equal groups with 6 in each, or an equivalent diagram

Written: 18 ÷ 3 = 6; 6 crackers

GRADING NOTE Grade both: the drawing of 3 equal groups, and the equation with the answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 7 • MAIN • Q2/5

A drawing shows 3 rows with 6 stars in each row.

Which equation matches the drawing?

A. 3 + 6 = ?

B. 6 ÷ 3 = ?

C. 3 × 6 = ?

D. 6 − 3 = ?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 7 • MAIN • Q3/5

197

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3 × 6 = ?

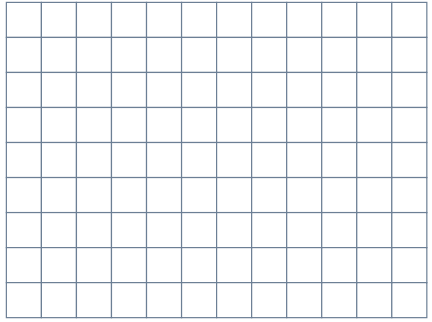
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 7 • MAIN • Q3/5

A 20-foot board is cut into pieces that are each 4 feet long.

Draw a diagram. Then write an equation with a symbol for the unknown and solve it.



Equation: _____

Pieces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 7 • MAIN • Q4/5

198

QUESTION ID

S7-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Any diagram that shows a total length of 20 feet made of 5 equal 4-foot pieces: for example a bar or number line marked in 4-unit parts, or 5 pieces each labeled 4 ft with the total of 20 ft shown

Written: 20 ÷ 4 = p; 5 pieces

GRADING NOTE Any equation with a symbol for the unknown (such as p × 4 = 20) is correct. Grade both the diagram and the equation with its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 7 • MAIN • Q4/5

True or false?

A drawing of 5 groups with 3 dots in each group matches the equation 5 + 3 = 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 7 • MAIN • Q5/5

199

QUESTION ID

S7-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 7 • MAIN • Q5/5

SET 7 • 3.OA.A.3

BACKWARD 1

Use an equation with a symbol for the unknown in a one-step word problem • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Lena has 23 shells. She finds some more and now has 41.

Write an equation with a symbol for the unknown. Then solve it.

Equation: _____

Shells found: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 7 • BACKWARD 1 • Q1/5

201

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

23 + ? = 41; 18

GRADING NOTE Any equation with a symbol for the unknown (such as 41 − 23 = s) is correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 7 • BACKWARD 1 • Q1/5

True or false?

The equation 52 − b = 30 matches "52 birds were on a wire. Some flew away, and 30 were left."

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 7 • BACKWARD 1 • Q2/5

202

QUESTION ID

S7-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 7 • BACKWARD 1 • Q2/5

A box had some pens. 14 more were put in, and now there are 50.

Which equation matches?

A. p + 14 = 50

B. p − 14 = 50

C. 14 − p = 50

D. 50 + 14 = p

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 7 • BACKWARD 1 • Q3/5

203

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. p + 14 = 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 7 • BACKWARD 1 • Q3/5

A rope is 60 inches long. Kim cuts off 25 inches.

Write an equation with a symbol for the length left. Then solve it.

Equation: _____

Length left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 7 • BACKWARD 1 • Q4/5

204

QUESTION ID

S7-B1-Q4

TYPE

Short answer

CORRECT ANSWER

60 − 25 = r; 35 inches

GRADING NOTE Any equation with a symbol for the unknown is correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 7 • BACKWARD 1 • Q4/5

True or false?

The equation 18 + 9 = n matches "There were 18 cats. 9 cats left. How many are still here?"

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 7 • BACKWARD 1 • Q5/5

205

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

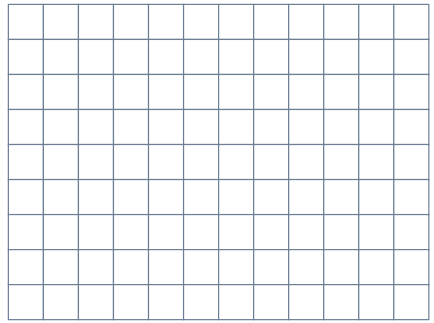
Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 3.OA.A.3

BACKWARD 2

Draw arrays with up to 5 rows and 5 columns • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

Draw an array with 3 rows of 4 dots.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 7 • BACKWARD 2 • Q1/5

207

QUESTION ID

S7-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

An array of 3 rows with 4 dots in each row (12 dots)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 7 • BACKWARD 2 • Q1/5

True or false?

An array with 4 rows of 2 dots shows $2 + 2 + 2 + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 7 • BACKWARD 2 • Q2/5

208

QUESTION ID

S7-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 7 • BACKWARD 2 • Q2/5

How many dots do you draw for an array with 5 rows of 3 dots?

A. 53

B. 8

C. 15

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 7 • BACKWARD 2 • Q3/5

209

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

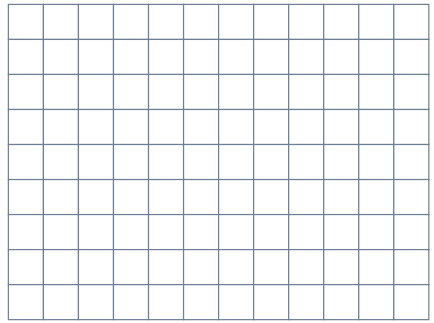
C. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 7 • BACKWARD 2 • Q3/5

Draw an array with 2 rows of 5 dots.

Then write an addition equation for it.



Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 7 • BACKWARD 2 • Q4/5

210

QUESTION ID

S7-B2-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: 2 rows with 5 dots in each row

Written: 5 + 5 = 10

GRADING NOTE Grade both: the array, and the equation. 2 + 2 + 2 + 2 + 2 = 10 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 7 • BACKWARD 2 • Q4/5

True or false?

An array with 3 rows of 3 dots has 6 dots.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 7 • BACKWARD 2 • Q5/5

211

QUESTION ID

S7-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 7 • BACKWARD 2 • Q5/5

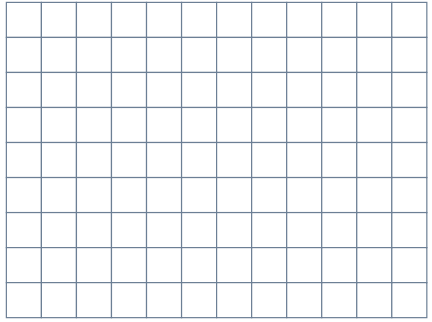
SET 7 • 3.OA.A.3

FORWARD 1

Use a drawing and an equation to solve a multiplicative comparison problem • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A blue scarf is 3 feet long. A red scarf is 4 times as long.

Draw a tape diagram. Then write an equation and find the red scarf's length.



Equation: _____

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 7 • FORWARD 1 • Q1/5

213

QUESTION ID

S7-F1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A tape for the blue scarf with 1 part of 3 feet, and a tape for the red scarf with 4 equal parts of 3 feet

Written: r = 4 × 3; 12 feet

GRADING NOTE Grade both: the tape diagram, and the equation with the answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 7 • FORWARD 1 • Q1/5

True or false?

The equation 36 = 4 × n matches "36 is 4 times as many as n."

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 7 • FORWARD 1 • Q2/5

214

QUESTION ID

S7-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 7 • FORWARD 1 • Q2/5

A school has 8 teachers and 5 times as many parent helpers.

Which equation finds the number of parent helpers, p?

A. p = 8 − 5

B. p = 5 + 8

C. p = 5 × 8

D. p = 8 ÷ 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 7 • FORWARD 1 • Q3/5

215

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. p = 5 × 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 7 • FORWARD 1 • Q3/5

Ty swam 30 laps. That is 6 times as many laps as Al swam.

Write an equation with a symbol for Al's laps. Then solve it.

Equation: _____

Laps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 7 • FORWARD 1 • Q4/5

216

QUESTION ID

S7-F1-Q4

TYPE

Short answer

CORRECT ANSWER

6 × a = 30; 5 laps

GRADING NOTE Any equivalent equation (such as a = 30 ÷ 6) is correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 7 • FORWARD 1 • Q4/5

True or false?

"21 is 3 times as many as 7" can be written as $21 = 3 + 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 7 • FORWARD 1 • Q5/5

217

QUESTION ID

S7-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 7 • FORWARD 1 • Q5/5

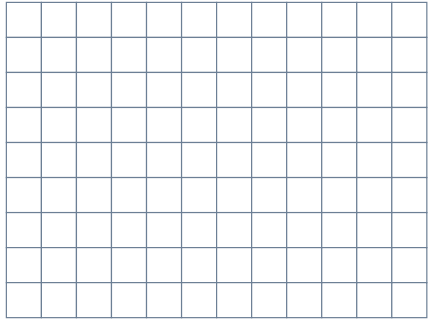
SET 7 • 3.OA.A.3

FORWARD 2

Use a tape diagram to find a fraction of a fractional or mixed-number quantity in context • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A path is $\frac{3}{4}$ mile long. Kai walked $\frac{2}{3}$ of it.

Draw a tape diagram. Then find how far Kai walked.



Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 7 • FORWARD 2 • Q1/5

219

QUESTION ID

S7-F2-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A tape for 3/4 mile split into 3 equal parts of 1/4 mile, with 2 parts marked (or another diagram showing 2/3 of 3/4)

Written: 1/2 mile

GRADING NOTE 2/4 mile is also correct. Grade both: the diagram, and the distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 7 • FORWARD 2 • Q1/5

True or false?

$\frac{3}{4}$ of a $\frac{2}{3}$-pound bag of rice is $\frac{1}{2}$ pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 7 • FORWARD 2 • Q2/5

220

QUESTION ID

S7-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 7 • FORWARD 2 • Q2/5

A recipe uses $1\frac{1}{2}$ cups of milk. Leo makes $\frac{1}{2}$ of the recipe.

How much milk does he use?

A. $\frac{3}{4}$ cup

B. 2 cups

C. 3 cups

D. 1 cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 7 • FORWARD 2 • Q3/5

221

QUESTION ID

S7-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3/4 cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 7 • FORWARD 2 • Q3/5

A rope is $2\frac{1}{2}$ feet long. $\frac{3}{5}$ of it is red.

How many feet of the rope are red?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 7 • FORWARD 2 • Q4/5

222

QUESTION ID

S7-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1 1/2 feet

GRADING NOTE 3/2 feet is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 7 • FORWARD 2 • Q4/5

True or false?

$\frac{2}{5}$ of a $\frac{5}{6}$-liter bottle of juice is $\frac{1}{2}$ liter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 7 • FORWARD 2 • Q5/5

223

QUESTION ID

S7-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 7 • FORWARD 2 • Q5/5

SET 8 • 3.OA.A.4 • OPERATIONS & ALGEBRAIC THINKING

Find the unknown whole number in a multiplication equation

MAIN	Grade 3 main question	**3.OA.A.4**
BACKWARD 1	Find the unknown number in an addition or subtraction equation w…	**1.OA.D.8**
BACKWARD 2	Skip-count to find a number in a counting sequence	**2.NBT.A.2**
FORWARD 1	Find factor pairs and decide whether a number is a factor	**4.OA.B.4**
FORWARD 2	Find an unknown factor with a two-digit factor	**5.NBT.B.6**

SET 8 • 3.OA.A.4

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4

Find the unknown number.

$7 \times ? = 56$

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 8 • MAIN • Q1/5

226

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 8 • MAIN • Q1/5

Find the unknown number.

? × 4 = 36

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 8 • MAIN • Q2/5

227

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 8 • MAIN • Q2/5

Which number makes the equation true?

$6 \times n = 42$

A. 36

B. 7

C. 48

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 8 • MAIN • Q3/5

228

QUESTION ID

S8-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 8 • MAIN • Q3/5

True or false?

In 4 × ▢ = 32, the unknown number is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 8 • MAIN • Q4/5

229

QUESTION ID

S8-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 8 • MAIN • Q4/5

Find the unknown number.

63 = 9 × ?

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 8 • MAIN • Q5/5

230

QUESTION ID

S8-M-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 8 • MAIN • Q5/5

SET 8 • 3.OA.A.4

BACKWARD 1

Find the unknown number in an addition or subtraction equation within 20 • Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8

Find the unknown number.

9 + ? = 15

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 8 • BACKWARD 1 • Q1/5

232

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 8 • BACKWARD 1 • Q1/5

True or false?

In 13 − ? = 6, the unknown number is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 8 • BACKWARD 1 • Q2/5

233

QUESTION ID

S8-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 8 • BACKWARD 1 • Q2/5

? + 7 = 16

A. 10

B. 9

C. 23

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 8 • BACKWARD 1 • Q3/5

234

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 8 • BACKWARD 1 • Q3/5

Find the unknown number.

14 = ? + 5

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 8 • BACKWARD 1 • Q4/5

235

QUESTION ID

S8-B1-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 8 • BACKWARD 1 • Q4/5

True or false?

In ? − 4 = 9, the unknown number is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 8 • BACKWARD 1 • Q5/5

236

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 3.OA.A.4

BACKWARD 2

Skip-count to find a number in a counting sequence • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2

Skip-count by 5s from 5.

How many numbers do you say to reach 35?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 8 • BACKWARD 2 • Q1/5

238

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 8 • BACKWARD 2 • Q1/5

True or false?

Skip-counting by 10s from 10, the 4th number is 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 8 • BACKWARD 2 • Q2/5

239

QUESTION ID

S8-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 8 • BACKWARD 2 • Q2/5

Skip-count by 5s from 5.

What is the 6th number?

A. 11

B. 35

C. 25

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 8 • BACKWARD 2 • Q3/5

240

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 8 • BACKWARD 2 • Q3/5

Skip-count by 100s from 100.

What is the 5th number?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 8 • BACKWARD 2 • Q4/5

241

QUESTION ID

S8-B2-Q4

TYPE

Short answer

CORRECT ANSWER

500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 8 • BACKWARD 2 • Q4/5

True or false?

Skip-counting by 10s from 10, the 7th number is 80.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 8 • BACKWARD 2 • Q5/5

242

QUESTION ID

S8-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 3.OA.A.4

FORWARD 1

Find factor pairs and decide whether a number is a factor • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 20.

Factor pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • FORWARD 1 • Q1/5

244

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1 × 20, 2 × 10, 4 × 5

GRADING NOTE All three pairs are required, in any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • FORWARD 1 • Q1/5

True or false?

7 is a factor of 63.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • FORWARD 1 • Q2/5

245

QUESTION ID

S8-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • FORWARD 1 • Q2/5

Which is a factor pair of 36?

A. 5 × 7

B. 4 × 9

C. 3 × 11

D. 6 × 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • FORWARD 1 • Q3/5

246

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4 × 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • FORWARD 1 • Q3/5

Is 6 a factor of 44? Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • FORWARD 1 • Q4/5

247

QUESTION ID

S8-F1-Q4

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE No whole number times 6 equals 44 (44 ÷ 6 = 7 R 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • FORWARD 1 • Q4/5

True or false?

8 is a factor of 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • FORWARD 1 • Q5/5

248

QUESTION ID

S8-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • 3.OA.A.4

FORWARD 2

Find an unknown factor with a two-digit factor • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

Find the unknown number.

$24 \times ? = 912$

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 8 • FORWARD 2 • Q1/5

250

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

38

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 8 • FORWARD 2 • Q1/5

True or false?

If 15 × n = 645, then n = 43.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 8 • FORWARD 2 • Q2/5

251

QUESTION ID

S8-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 8 • FORWARD 2 • Q2/5

? × 32 = 1,248

A. 41

B. 1,280

C. 38

D. 39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 8 • FORWARD 2 • Q3/5

252

QUESTION ID

S8-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 8 • FORWARD 2 • Q3/5

Find the unknown number.

2,100 = 75 × ?

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 8 • FORWARD 2 • Q4/5

253

QUESTION ID

S8-F2-Q4

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 8 • FORWARD 2 • Q4/5

True or false?

If 18 × n = 864, then n = 46.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 8 • FORWARD 2 • Q5/5

254

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 8 • FORWARD 2 • Q5/5

SET 9 • 3.OA.A.4 • OPERATIONS & ALGEBRAIC THINKING

Find the unknown whole number in a division equation

MAIN	Grade 3 main question	**3.OA.A.4**
BACKWARD 1	Find the unknown number in a subtraction or addition equation w…	**1.OA.D.8**
BACKWARD 2	Add and subtract within 20 mentally	**2.OA.B.2**
FORWARD 1	Find the unknown number in a division equation with a larger divi…	**4.NBT.B.6**
FORWARD 2	Find the unknown number in a division equation with a two-digit d…	**5.NBT.B.6**

SET 9 • 3.OA.A.4

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4

Find the unknown number.

45 ÷ ? = 9

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 9 • MAIN • Q1/5

257

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 9 • MAIN • Q1/5

Find the unknown number.

? ÷ 6 = 7

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 9 • MAIN • Q2/5

258

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 9 • MAIN • Q2/5

Which number makes the equation true?

72 ÷ n = 8

A. 8

B. 64

C. 80

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 9 • MAIN • Q3/5

259

QUESTION ID

S9-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 9 • MAIN • Q3/5

True or false?

In ? ÷ 4 = 8, the unknown number is 32.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 9 • MAIN • Q4/5

260

QUESTION ID

S9-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 9 • MAIN • Q4/5

True or false?

In 49 ÷ ▢ = 7, the unknown number is 42.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 9 • MAIN • Q5/5

261

QUESTION ID

S9-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 9 • MAIN • Q5/5

SET 9 • 3.OA.A.4

BACKWARD 1

Find the unknown number in a subtraction or addition equation within 20 • Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8

Find the unknown number.

17 − ? = 9

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 9 • BACKWARD 1 • Q1/5

263

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 9 • BACKWARD 1 • Q1/5

True or false?

In ? + 6 = 14, the unknown number is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 9 • BACKWARD 1 • Q2/5

264

QUESTION ID

S9-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 9 • BACKWARD 1 • Q2/5

$12 = ? - 3$

A. 15

B. 14

C. 4

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 9 • BACKWARD 1 • Q3/5

265

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 9 • BACKWARD 1 • Q3/5

Find the unknown number.

? − 7 = 7

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 9 • BACKWARD 1 • Q4/5

266

QUESTION ID

S9-B1-Q4

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 9 • BACKWARD 1 • Q4/5

True or false?

In 16 − ? = 10, the unknown number is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 9 • BACKWARD 1 • Q5/5

267

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 3.OA.A.4

BACKWARD 2

Add and subtract within 20 mentally • Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2

Subtract.

15 − 8

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 9 • BACKWARD 2 • Q1/5

269

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 9 • BACKWARD 2 • Q1/5

True or false?

9 + 8 = 17

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 9 • BACKWARD 2 • Q2/5

270

QUESTION ID

S9-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 9 • BACKWARD 2 • Q2/5

13 − 6 = ?

A. 6

B. 7

C. 19

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 9 • BACKWARD 2 • Q3/5

271

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 9 • BACKWARD 2 • Q3/5

Add.

7 + 6

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 9 • BACKWARD 2 • Q4/5

272

QUESTION ID

S9-B2-Q4

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 9 • BACKWARD 2 • Q4/5

True or false?

$16 - 9 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 9 • BACKWARD 2 • Q5/5

273

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 3.OA.A.4

FORWARD 1

Find the unknown number in a division equation with a larger dividend • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Find the unknown number.

? ÷ 4 = 125

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 9 • FORWARD 1 • Q1/5

275

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 9 • FORWARD 1 • Q1/5

True or false?

In 312 ÷ n = 6, n = 52.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 9 • FORWARD 1 • Q2/5

276

QUESTION ID

S9-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 9 • FORWARD 1 • Q2/5

$848 \div ? = 8$

A. 16

B. 856

C. 840

D. 106

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 9 • FORWARD 1 • Q3/5

277

QUESTION ID

S9-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 106

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 9 • FORWARD 1 • Q3/5

Find the unknown number.

? ÷ 7 = 36

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 9 • FORWARD 1 • Q4/5

278

QUESTION ID

S9-F1-Q4

TYPE

Short answer

CORRECT ANSWER

252

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 9 • FORWARD 1 • Q4/5

True or false?

In 450 ÷ n = 5, n = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 9 • FORWARD 1 • Q5/5

279

QUESTION ID

S9-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 9 • FORWARD 1 • Q5/5

SET 9 • 3.OA.A.4

FORWARD 2

Find the unknown number in a division equation with a two-digit divisor • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

Find the unknown number.

? ÷ 25 = 36

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 9 • FORWARD 2 • Q1/5

281

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

900

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 9 • FORWARD 2 • Q1/5

True or false?

In 1,092 ÷ n = 42, n = 26.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 9 • FORWARD 2 • Q2/5

282

QUESTION ID

S9-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 9 • FORWARD 2 • Q2/5

1,512 ÷ ? = 36

A. 1,476

B. 54

C. 42

D. 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 9 • FORWARD 2 • Q3/5

283

QUESTION ID

S9-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 9 • FORWARD 2 • Q3/5

Find the unknown number.

? ÷ 16 = 45

Unknown: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 9 • FORWARD 2 • Q4/5

284

QUESTION ID

S9-F2-Q4

TYPE

Short answer

CORRECT ANSWER

720

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 9 • FORWARD 2 • Q4/5

True or false?

In 2,000 ÷ n = 80, n = 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 9 • FORWARD 2 • Q5/5

285

QUESTION ID

S9-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 9 • FORWARD 2 • Q5/5

SET 10 • 3.OA.B.5 • OPERATIONS & ALGEBRAIC THINKING

Use the commutative property of multiplication

MAIN	Grade 3 main question	**3.OA.B.5**
BACKWARD 1	Add in either order	**1.OA.B.3**
BACKWARD 2	Find the total of an array by rows or by columns	**2.OA.C.4**
FORWARD 1	Use the commutative property with multi-digit products	**4.NBT.B.5**
FORWARD 2	Multiply two fractions in either order	**5.NF.B.4.a**

SET 10 • 3.OA.B.5

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

$7 \times 8 = 56$.

Without multiplying, what is 8×7? Explain.

Product: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 10 • MAIN • Q1/5

288

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

56

GRADING NOTE Changing the order of the factors does not change the product (the commutative property). Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 10 • MAIN • Q1/5

Which is equal to 9 × 3?

A. 3 × 9

B. 93

C. 9 × 9

D. 9 + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 10 • MAIN • Q2/5

289

QUESTION ID

S10-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 3 × 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 10 • MAIN • Q2/5

True or false?

$5 \times 9 = 9 \times 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 10 • MAIN • Q3/5

290

QUESTION ID

S10-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 10 • MAIN • Q3/5

An array has 4 rows of 7 dots. It is turned on its side, so it now has 7 rows.

How many dots are in each row now? How many dots are there in all?

Dots in each row: _____

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 10 • MAIN • Q4/5

291

QUESTION ID

S10-M-Q4

TYPE

Short answer

CORRECT ANSWER

4; 28

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 10 • MAIN • Q4/5

True or false?

$8 \times 4 = 4 + 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 10 • MAIN • Q5/5

292

QUESTION ID

S10-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 10 • MAIN • Q5/5

SET 10 • 3.OA.B.5

BACKWARD 1

Add in either order • Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3

9 + 5 = 14.

What is 5 + 9?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 10 • BACKWARD 1 • Q1/5

294

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 10 • BACKWARD 1 • Q1/5

True or false?

$7 + 6 = 6 + 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 10 • BACKWARD 1 • Q2/5

295

QUESTION ID

S10-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 10 • BACKWARD 1 • Q2/5

Which is equal to 4 + 9?

A. 49

B. 4 + 4

C. 9 + 4

D. 9 − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 10 • BACKWARD 1 • Q3/5

296

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 9 + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 10 • BACKWARD 1 • Q3/5

Fill in the blank.

6 + 8 = 8 + ___

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 10 • BACKWARD 1 • Q4/5

297

QUESTION ID

S10-B1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 10 • BACKWARD 1 • Q4/5

True or false?

$5 + 2 = 5 - 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 10 • BACKWARD 1 • Q5/5

298

QUESTION ID

S10-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 10 • BACKWARD 1 • Q5/5

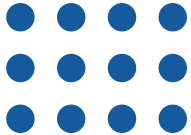
SET 10 • 3.OA.B.5

BACKWARD 2

Find the total of an array by rows or by columns • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

Add the rows. Then add the columns.

What total do you get each way?



By rows: _____

By columns: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 10 • BACKWARD 2 • Q1/5

300

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12; 12

GRADING NOTE 4 + 4 + 4 = 12 and 3 + 3 + 3 + 3 = 12. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 10 • BACKWARD 2 • Q1/5

True or false?

An array with 2 rows of 5 dots has the same number of dots as an array with 5 rows of 2 dots.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 10 • BACKWARD 2 • Q2/5

301

QUESTION ID

S10-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 10 • BACKWARD 2 • Q2/5

An array has 3 rows of 5 dots.

Which addition adds its columns?

A. 5 + 5 + 5

B. 5 + 3 + 5

C. 3 + 5

D. 3 + 3 + 3 + 3 + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 10 • BACKWARD 2 • Q3/5

302

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 + 3 + 3 + 3 + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 10 • BACKWARD 2 • Q3/5

How many dots are in an array with 4 rows of 5 dots?

Dots: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 10 • BACKWARD 2 • Q4/5

303

QUESTION ID

S10-B2-Q4

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 10 • BACKWARD 2 • Q4/5

True or false?

An array with 4 rows of 3 dots has more dots than an array with 3 rows of 4 dots.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 10 • BACKWARD 2 • Q5/5

304

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 3.OA.B.5

FORWARD 1

Use the commutative property with multi-digit products • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

True or false?

$24 \times 15 = 15 \times 24$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 10 • FORWARD 1 • Q1/5

306

QUESTION ID

S10-F1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 10 • FORWARD 1 • Q1/5

36 × 7 = 252.

What is 7 × 36?

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 10 • FORWARD 1 • Q2/5

307

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

252

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 10 • FORWARD 1 • Q2/5

Which is equal to 48 × 25?

A. 25 × 48

B. 48 × 52

C. 48 + 25

D. 84 × 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 10 • FORWARD 1 • Q3/5

308

QUESTION ID

S10-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 25 × 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 10 • FORWARD 1 • Q3/5

Is 19 × 6 equal to 6 × 19? Find the product.

Equal? _____

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 10 • FORWARD 1 • Q4/5

309

QUESTION ID

S10-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes; 114

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 10 • FORWARD 1 • Q4/5

True or false?

$125 \times 8 = 8 \times 152$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 10 • FORWARD 1 • Q5/5

310

QUESTION ID

S10-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 10 • FORWARD 1 • Q5/5

SET 10 • 3.OA.B.5

FORWARD 2

Multiply two fractions in either order • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

True or false?

$\frac{2}{3} \times \frac{4}{5} = \frac{4}{5} \times \frac{2}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 10 • FORWARD 2 • Q1/5

312

QUESTION ID

S10-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 10 • FORWARD 2 • Q1/5

Multiply.

$\frac{3}{4} \times \frac{2}{5}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 10 • FORWARD 2 • Q2/5

313

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3/10

GRADING NOTE 6/20 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 10 • FORWARD 2 • Q2/5

Which is equal to $\frac{5}{6} \times \frac{3}{8}$?

A. $\frac{8}{3} \times \frac{5}{6}$

B. $\frac{5}{6} + \frac{3}{8}$

C. $\frac{3}{8} \times \frac{5}{6}$

D. $\frac{6}{5} \times \frac{3}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 10 • FORWARD 2 • Q3/5

314

QUESTION ID

S10-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/8 × (5/6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 10 • FORWARD 2 • Q3/5

$\frac{3}{5} \times \frac{5}{6} = \frac{1}{2}$.

What is $\frac{5}{6} \times \frac{3}{5}$?

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 10 • FORWARD 2 • Q4/5

315

QUESTION ID

S10-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 10 • FORWARD 2 • Q4/5

True or false?

$\frac{1}{2} \times \frac{3}{4} = \frac{3}{4} \times 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 10 • FORWARD 2 • Q5/5

316

QUESTION ID

S10-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 10 • FORWARD 2 • Q5/5

SET 11 • 3.OA.B.5 • OPERATIONS & ALGEBRAIC THINKING

Use the associative property of multiplication

MAIN	Grade 3 main question	**3.OA.B.5**
BACKWARD 1	Add three numbers in any grouping	**1.OA.B.3**
BACKWARD 2	Add within 20 mentally	**2.OA.B.2**
FORWARD 1	Use grouping and multiples of 10 to multiply	**4.NBT.B.5**
FORWARD 2	Multiply by powers of 10 written with exponents	**5.NBT.A.2**

SET 11 • 3.OA.B.5

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

Find the product by first multiplying 2 × 5.

2 × 5 × 7

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 11 • MAIN • Q1/5

319

QUESTION ID

S11-M-Q1

TYPE

Short answer

CORRECT ANSWER

70

GRADING NOTE 2 × 5 = 10 and 10 × 7 = 70.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 11 • MAIN • Q1/5

Which is equal to (4 × 2) × 3?

A. 4 + (2 × 3)

B. 4 × 2 + 3

C. 4 × (2 × 3)

D. (4 + 2) × 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 11 • MAIN • Q2/5

320

QUESTION ID

S11-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 4 × (2 × 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 11 • MAIN • Q2/5

True or false?

$(3 \times 2) \times 4 = 3 \times (2 \times 4)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 11 • MAIN • Q3/5

321

QUESTION ID

S11-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 11 • MAIN • Q3/5

Show two different ways to group 5 × 2 × 6. Then find the product.

Way 1: _____

Way 2: _____

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 11 • MAIN • Q4/5

322

QUESTION ID

S11-M-Q4

TYPE

Short answer

CORRECT ANSWER

(5 × 2) × 6 = 60;
5 × (2 × 6) = 60; 60

GRADING NOTE Any two different groupings with the product 60 are correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 11 • MAIN • Q4/5

True or false?

$(2 \times 4) \times 5 = 2 + (4 \times 5)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 11 • MAIN • Q5/5

323

QUESTION ID

S11-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 11 • MAIN • Q5/5

SET 11 • 3.OA.B.5

BACKWARD 1

Add three numbers in any grouping • Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3

Add.

7 + 3 + 5

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 11 • BACKWARD 1 • Q1/5

325

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 11 • BACKWARD 1 • Q1/5

True or false?

$(4 + 6) + 3 = 4 + (6 + 3)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 11 • BACKWARD 1 • Q2/5

326

QUESTION ID

S11-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 11 • BACKWARD 1 • Q2/5

Which sum is equal to 8 + 5 + 2?

A. 10 + 5

B. 8 + 7 + 2

C. 10 + 7

D. 13 + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 11 • BACKWARD 1 • Q3/5

327

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10 + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 11 • BACKWARD 1 • Q3/5

Add by making a ten first.

6 + 9 + 4

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 11 • BACKWARD 1 • Q4/5

328

QUESTION ID

S11-B1-Q4

TYPE

Short answer

CORRECT ANSWER

19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 11 • BACKWARD 1 • Q4/5

True or false?

$(5 + 5) + 4 = 10 + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 11 • BACKWARD 1 • Q5/5

329

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 3.OA.B.5

BACKWARD 2

Add within 20 mentally • Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2

Add.

8 + 9

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 11 • BACKWARD 2 • Q1/5

331

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 11 • BACKWARD 2 • Q1/5

True or false?

6 + 7 = 13

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 11 • BACKWARD 2 • Q2/5

332

QUESTION ID

S11-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 11 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

9 + 9 = ?

A. 81

B. 17

C. 19

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 11 • BACKWARD 2 • Q3/5

333

QUESTION ID

S11-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 11 • BACKWARD 2 • Q3/5

Add.

10 + 7

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 11 • BACKWARD 2 • Q4/5

334

QUESTION ID

S11-B2-Q4

TYPE

Short answer

CORRECT ANSWER

17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 11 • BACKWARD 2 • Q4/5

True or false?

5 + 8 = 12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 11 • BACKWARD 2 • Q5/5

335

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 3.OA.B.5

FORWARD 1

Use grouping and multiples of 10 to multiply • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Find 4 × 5 × 13 by first multiplying 4 × 5.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 11 • FORWARD 1 • Q1/5

337

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

260

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 11 • FORWARD 1 • Q1/5

True or false?

$6 \times 25 \times 4 = 6 \times 100$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 11 • FORWARD 1 • Q2/5

338

QUESTION ID

S11-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 11 • FORWARD 1 • Q2/5

$2 \times 37 \times 5 = 37 \times ?$

A. 2

B. 74

C. 10

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 11 • FORWARD 1 • Q3/5

339

QUESTION ID

S11-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 11 • FORWARD 1 • Q3/5

Multiply.

$3 \times 20 \times 5$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 11 • FORWARD 1 • Q4/5

340

QUESTION ID

S11-F1-Q4

TYPE

Short answer

CORRECT ANSWER

300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 11 • FORWARD 1 • Q4/5

True or false?

$25 \times 3 \times 4 = 75 \times 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 11 • FORWARD 1 • Q5/5

341

QUESTION ID

S11-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 11 • FORWARD 1 • Q5/5

SET 11 • 3.OA.B.5

FORWARD 2

Multiply by powers of 10 written with exponents • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

4×10^2

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • FORWARD 2 • Q1/5

343

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • FORWARD 2 • Q1/5

True or false?

$7 \times 10 \times 10 \times 10 = 7 \times 10^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • FORWARD 2 • Q2/5

344

QUESTION ID

S11-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • FORWARD 2 • Q2/5

Which is equal to 36 × 100?

A. 36 + 100

B. 36×10^3

C. 36×10^2

D. 36 × 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • FORWARD 2 • Q3/5

345

QUESTION ID

S11-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 36×10^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • FORWARD 2 • Q3/5

Find $25 \times 4 \times 10^2$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • FORWARD 2 • Q4/5

346

QUESTION ID

S11-F2-Q4

TYPE

Short answer

CORRECT ANSWER

10,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • FORWARD 2 • Q4/5

True or false?

$3 \times 10^2 = 3{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • FORWARD 2 • Q5/5

347

QUESTION ID

S11-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • FORWARD 2 • Q5/5

SET 12 • 3.OA.B.5 • OPERATIONS & ALGEBRAIC THINKING

Use the distributive property to multiply

MAIN	Grade 3 main question	**3.OA.B.5**
BACKWARD 1	Add two-digit numbers within 100	**2.NBT.B.5**
BACKWARD 2	Split an array into two parts and add the parts	**2.OA.C.4**
FORWARD 1	Multiply a two-digit number by a one-digit number with the distrib…	**4.NBT.B.5**
FORWARD 2	Evaluate expressions with parentheses that use the distributive pr…	**5.OA.A.1**

SET 12 • 3.OA.B.5

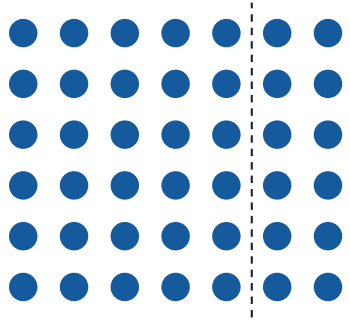
MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

ANSWER KEY Reduced copy of the question page, with its answer

The dashed line splits the array of 6 rows of 7 into 6 rows of 5 and 6 rows of 2.

Use 6 × 7 = 6 × 5 + 6 × 2 to find 6 × 7.



Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 12 • MAIN • Q1/5

350

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

42

GRADING NOTE 6 × 5 = 30, 6 × 2 = 12, and 30 + 12 = 42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 12 • MAIN • Q1/5

Which is equal to 4 × 9?

A. 4 × 5 + 4

B. 4 × 5 × 4

C. 4 + 9

D. 4 × 5 + 4 × 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 12 • MAIN • Q2/5

351

QUESTION ID

S12-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 4 × 5 + 4 × 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 12 • MAIN • Q2/5

True or false?

$9 \times 6 = 9 \times 3 + 9 \times 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 12 • MAIN • Q3/5

352

QUESTION ID

S12-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 12 • MAIN • Q3/5

Use the distributive property to find 8 × 6. Show the parts you add.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 12 • MAIN • Q4/5

353

QUESTION ID

S12-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

48

GRADING NOTE REQUIRED WORK: Any correct break-apart, for example 8 × 6 = 8 × 5 + 8 × 1 = 40 + 8, or 8 × 6 = 4 × 6 + 4 × 6 = 24 + 24. The product without a break-apart earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 12 • MAIN • Q4/5

True or false?

$7 \times 8 = 7 \times 4 + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 12 • MAIN • Q5/5

354

QUESTION ID

S12-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 12 • MAIN • Q5/5

SET 12 • 3.OA.B.5

BACKWARD 1

Add two-digit numbers within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

40 + 16

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 12 • BACKWARD 1 • Q1/5

356

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 12 • BACKWARD 1 • Q1/5

True or false?

35 + 21 = 56

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 12 • BACKWARD 1 • Q2/5

357

QUESTION ID

S12-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 12 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

30 + 18 = ?

A. 38

B. 12

C. 48

D. 58

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 12 • BACKWARD 1 • Q3/5

358

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 12 • BACKWARD 1 • Q3/5

Add.

26 + 26

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 12 • BACKWARD 1 • Q4/5

359

QUESTION ID

S12-B1-Q4

TYPE

Short answer

CORRECT ANSWER

52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 12 • BACKWARD 1 • Q4/5

True or false?

45 + 27 = 62

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 12 • BACKWARD 1 • Q5/5

360

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 3.OA.B.5

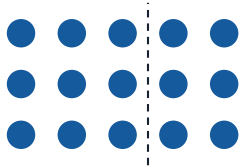
BACKWARD 2

Split an array into two parts and add the parts • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

ANSWER KEY Reduced copy of the question page, with its answer

The dashed line splits the array.

How many dots are on each side?
How many dots are there in all?



Left: _____

Right: _____

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 12 • BACKWARD 2 • Q1/5

362

QUESTION ID
S12-B2-Q1

TYPE
Short answer

CORRECT ANSWER
9; 6; 15

GRADING NOTE All three parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 12 • BACKWARD 2 • Q1/5

True or false?

An array of 4 rows of 5 dots can be split into 4 rows of 3 dots and 4 rows of 2 dots.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 12 • BACKWARD 2 • Q2/5

363

QUESTION ID

S12-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 12 • BACKWARD 2 • Q2/5

An array of 2 rows of 5 dots is split into 2 rows of 4 dots and 2 rows of 1 dot.

How many dots are there in all?

A. 10

B. 8

C. 12

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 12 • BACKWARD 2 • Q3/5

364

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 12 • BACKWARD 2 • Q3/5

An array of 5 rows of 4 dots is split into two arrays of 5 rows of 2 dots.

How many dots are in each part?

Dots in each part: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 12 • BACKWARD 2 • Q4/5

365

QUESTION ID

S12-B2-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 12 • BACKWARD 2 • Q4/5

True or false?

Splitting an array into two parts changes the total number of dots.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 12 • BACKWARD 2 • Q5/5

366

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

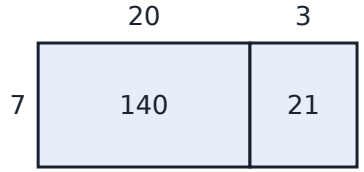
Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 3.OA.B.5

FORWARD 1

Multiply a two-digit number by a one-digit number with the distributive property • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Use the area model to find 7×23.



Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 12 • FORWARD 1 • Q1/5

368

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

161

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 12 • FORWARD 1 • Q1/5

True or false?

$6 \times 34 = 6 \times 30 + 6 \times 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 12 • FORWARD 1 • Q2/5

369

QUESTION ID

S12-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 12 • FORWARD 1 • Q2/5

Which is equal to 8 × 45?

A. 8 × 4 + 8 × 5

B. 8 × 40 + 8 × 5

C. 8 × 40 + 5

D. 8 + 40 + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 12 • FORWARD 1 • Q3/5

370

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8 × 40 + 8 × 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 12 • FORWARD 1 • Q3/5

Use the distributive property to find 5 × 68. Show your work.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 12 • FORWARD 1 • Q4/5

371

QUESTION ID

S12-F1-Q4

TYPE

Answer with required work

CORRECT ANSWER

340

GRADING NOTE REQUIRED WORK: Any correct break-apart, for example 5 × 60 + 5 × 8 = 300 + 40, or 5 × 70 − 5 × 2 = 350 − 10. The product without a break-apart earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 12 • FORWARD 1 • Q4/5

True or false?

$9 \times 52 = 450 + 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 12 • FORWARD 1 • Q5/5

372

QUESTION ID

S12-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 12 • FORWARD 1 • Q5/5

SET 12 • 3.OA.B.5

FORWARD 2

Evaluate expressions with parentheses that use the distributive property • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$6 \times (10 + 3)$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 12 • FORWARD 2 • Q1/5

374

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

78

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 12 • FORWARD 2 • Q1/5

True or false?

$4 \times (25 + 5) = 4 \times 25 + 4 \times 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 12 • FORWARD 2 • Q2/5

375

QUESTION ID

S12-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 12 • FORWARD 2 • Q2/5

Evaluate.

$(7 \times 20) + (7 \times 4)$

A. 168

B. 140

C. 31

D. 161

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 12 • FORWARD 2 • Q3/5

376

QUESTION ID

S12-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 168

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 12 • FORWARD 2 • Q3/5

Evaluate.

$3 \times (40 - 2)$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 12 • FORWARD 2 • Q4/5

377

QUESTION ID

S12-F2-Q4

TYPE

Short answer

CORRECT ANSWER

114

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 12 • FORWARD 2 • Q4/5

True or false?

$5 \times (8 + 2) = 42$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 12 • FORWARD 2 • Q5/5

378

QUESTION ID

S12-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 12 • FORWARD 2 • Q5/5

SET 13 • 3.OA.B.6 • OPERATIONS & ALGEBRAIC THINKING

Understand division as an unknown-factor problem

MAIN	Grade 3 main question	**3.OA.B.6**
BACKWARD 1	Understand subtraction as an unknown-addend problem	**1.OA.B.4**
BACKWARD 2	Write even numbers as the sum of two equal addends	**2.OA.C.3**
FORWARD 1	Use multiplication to find and check quotients	**4.NBT.B.6**
FORWARD 2	Divide a whole number by a unit fraction and check with multipli…	**5.NF.B.7.b**

SET 13 • 3.OA.B.6

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find 36 ÷ 9 by finding the unknown factor.

9 × ? = 36

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 13 • MAIN • Q1/5

381

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 13 • MAIN • Q1/5

Which multiplication fact helps you find 42 ÷ 6?

A. 42 − 6 = 36

B. 6 + 7 = 13

C. 6 × 7 = 42

D. 6 × 6 = 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 13 • MAIN • Q2/5

382

QUESTION ID

S13-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 6 × 7 = 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 13 • MAIN • Q2/5

Write the unknown-factor equation for 54 ÷ 9. Then find the quotient.

Equation: _____

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 13 • MAIN • Q3/5

383

QUESTION ID

S13-M-Q3

TYPE

Short answer

CORRECT ANSWER

9 × ? = 54; 6

GRADING NOTE ? × 9 = 54 is also correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 13 • MAIN • Q3/5

True or false?

56 ÷ 7 = 8 because 7 × 8 = 56.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 13 • MAIN • Q4/5

384

QUESTION ID

S13-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 13 • MAIN • Q4/5

True or false?

27 ÷ 3 = 8 because 3 × 8 = 27.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 13 • MAIN • Q5/5

385

QUESTION ID

S13-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 13 • MAIN • Q5/5

SET 13 • 3.OA.B.6

BACKWARD 1

Understand subtraction as an unknown-addend problem • Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4

Find 15 − 9 by thinking 9 + ? = 15.

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 13 • BACKWARD 1 • Q1/5

387

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 13 • BACKWARD 1 • Q1/5

True or false?

13 − 5 = 8 because 5 + 8 = 13.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 13 • BACKWARD 1 • Q2/5

388

QUESTION ID

S13-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 13 • BACKWARD 1 • Q2/5

Which addition fact helps you find 17 − 8?

A. 8 + 9 = 17

B. 8 + 8 = 16

C. 17 + 8 = 25

D. 9 − 8 = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 13 • BACKWARD 1 • Q3/5

389

QUESTION ID

S13-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8 + 9 = 17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 13 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the unknown-addend equation for 12 − 7. Then find the difference.

Equation: _____

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 13 • BACKWARD 1 • Q4/5

390

QUESTION ID

S13-B1-Q4

TYPE

Short answer

CORRECT ANSWER

7 + ? = 12; 5

GRADING NOTE ? + 7 = 12 is also correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 13 • BACKWARD 1 • Q4/5

True or false?

14 − 6 = 9 because 6 + 9 = 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 13 • BACKWARD 1 • Q5/5

391

QUESTION ID

S13-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 3.OA.B.6

BACKWARD 2

Write even numbers as the sum of two equal addends • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

ANSWER KEY Reduced copy of the question page, with its answer

Write 18 as the sum of two equal addends.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q1/5

393

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9 + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q1/5

True or false?

10 = 5 + 5, so 10 is even.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q2/5

394

QUESTION ID

S13-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q2/5

Which number can be written as the sum of two equal whole-number addends?

A. 17

B. 14

C. 13

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q3/5

395

QUESTION ID

S13-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q3/5

What number plus itself makes 12?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q4/5

396

QUESTION ID

S13-B2-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q4/5

True or false?

15 can be written as the sum of two equal whole-number addends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q5/5

397

QUESTION ID

S13-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 3.OA.B.6

FORWARD 1

Use multiplication to find and check quotients • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Find 84 ÷ 4 by thinking 4 × ? = 84.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 13 • FORWARD 1 • Q1/5

399

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 13 • FORWARD 1 • Q1/5

True or false?

96 ÷ 3 = 32 because 3 × 32 = 96.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 13 • FORWARD 1 • Q2/5

400

QUESTION ID

S13-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 13 • FORWARD 1 • Q2/5

Which equation checks 135 ÷ 5 = 27?

A. 135 × 5 = 675

B. 27 − 5 = 22

C. 5 × 27 = 135

D. 5 + 27 = 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 13 • FORWARD 1 • Q3/5

401

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 × 27 = 135

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 13 • FORWARD 1 • Q3/5

Write the unknown-factor equation for 224 ÷ 7. Then find the quotient.

Equation: _____

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 13 • FORWARD 1 • Q4/5

402

QUESTION ID

S13-F1-Q4

TYPE

Short answer

CORRECT ANSWER

7 × ? = 224; 32

GRADING NOTE ? × 7 = 224 is also correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 13 • FORWARD 1 • Q4/5

True or false?

152 ÷ 8 = 18 because 8 × 18 = 152.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 13 • FORWARD 1 • Q5/5

403

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 13 • FORWARD 1 • Q5/5

SET 13 • 3.OA.B.6

FORWARD 2

Divide a whole number by a unit fraction and check with multiplication • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b

Find $3 \div \frac{1}{4}$ by thinking $? \times \frac{1}{4} = 3$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b • Set 13 • FORWARD 2 • Q1/5

405

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b • Set 13 • FORWARD 2 • Q1/5

True or false?

$5 \div \frac{1}{2} = 10$ because $10 \times \frac{1}{2} = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b • Set 13 • FORWARD 2 • Q2/5

406

QUESTION ID

S13-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b • Set 13 • FORWARD 2 • Q2/5

Which equation checks $4 \div \frac{1}{3} = 12$?

A. $4 \times \frac{1}{3} = 12$

B. $12 \div 4 = 3$

C. $\frac{1}{3} + 12 = 4$

D. $12 \times \frac{1}{3} = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b • Set 13 • FORWARD 2 • Q3/5

407

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12 × (1/3) = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b • Set 13 • FORWARD 2 • Q3/5

Divide. Then write the multiplication equation that checks your answer.

$2 \div \frac{1}{6}$

Quotient: _____

Check: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b • Set 13 • FORWARD 2 • Q4/5

408

QUESTION ID

S13-F2-Q4

TYPE

Short answer

CORRECT ANSWER

12; 12 × 1/6 = 2

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b • Set 13 • FORWARD 2 • Q4/5

True or false?

$6 \div \frac{1}{3} = 2$ because $2 \times \frac{1}{3} = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b • Set 13 • FORWARD 2 • Q5/5

409

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b • Set 13 • FORWARD 2 • Q5/5

SET 14 • 3.OA.C.7 • OPERATIONS & ALGEBRAIC THINKING

Multiply fluently within 100

MAIN	Grade 3 main question	**3.OA.C.7**
BACKWARD 1	Add equal addends to find a total	**2.OA.C.4**
BACKWARD 2	Add doubles within 20	**2.OA.B.2**
FORWARD 1	Multiply a two-digit number by a one-digit number	**4.NBT.B.5**
FORWARD 2	Multiply multi-digit whole numbers with the standard algorithm	**5.NBT.B.5**

SET 14 • 3.OA.C.7

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

7 × 6

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 14 • MAIN • Q1/5

412

QUESTION ID

S14-M-Q1

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 14 • MAIN • Q1/5

Multiply.

9 × 8

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 14 • MAIN • Q2/5

413

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 14 • MAIN • Q2/5

$6 \times 8 = ?$

A. 42

B. 48

C. 54

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 14 • MAIN • Q3/5

414

QUESTION ID

S14-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 14 • MAIN • Q3/5

True or false?

$7 \times 7 = 49$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 14 • MAIN • Q4/5

415

QUESTION ID

S14-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 14 • MAIN • Q4/5

True or false?

$8 \times 4 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 14 • MAIN • Q5/5

QUESTION ID

S14-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 14 • MAIN • Q5/5

SET 14 • 3.OA.C.7

BACKWARD 1

Add equal addends to find a total • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

Add the equal addends.

4 + 4 + 4 + 4 + 4

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 14 • BACKWARD 1 • Q1/5

418

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 14 • BACKWARD 1 • Q1/5

True or false?

3 + 3 + 3 + 3 = 12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 14 • BACKWARD 1 • Q2/5

419

QUESTION ID

S14-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 14 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

5 + 5 + 5 = ?

A. 555

B. 15

C. 10

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 14 • BACKWARD 1 • Q3/5

420

QUESTION ID

S14-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 14 • BACKWARD 1 • Q3/5

An array has 4 rows of 3 dots.

How many dots are there?

Dots: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 14 • BACKWARD 1 • Q4/5

421

QUESTION ID

S14-B1-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 14 • BACKWARD 1 • Q4/5

True or false?

$2 + 2 + 2 + 2 + 2 = 12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 14 • BACKWARD 1 • Q5/5

422

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 3.OA.C.7

BACKWARD 2

Add doubles within 20 • Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2

Add.

7 + 7

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 14 • BACKWARD 2 • Q1/5

424

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 14 • BACKWARD 2 • Q1/5

True or false?

9 + 9 = 18

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 14 • BACKWARD 2 • Q2/5

425

QUESTION ID

S14-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 14 • BACKWARD 2 • Q2/5

8 + 8 = ?

A. 15

B. 88

C. 18

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 14 • BACKWARD 2 • Q3/5

426

QUESTION ID

S14-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 14 • BACKWARD 2 • Q3/5

Add.

6 + 6

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 14 • BACKWARD 2 • Q4/5

427

QUESTION ID

S14-B2-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 14 • BACKWARD 2 • Q4/5

True or false?

5 + 5 = 11

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 14 • BACKWARD 2 • Q5/5

428

QUESTION ID

S14-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 3.OA.C.7

FORWARD 1

Multiply a two-digit number by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Multiply.

7 × 46

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 14 • FORWARD 1 • Q1/5

430

QUESTION ID

S14-F1-Q1

TYPE

Short answer

CORRECT ANSWER

322

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 14 • FORWARD 1 • Q1/5

True or false?

$8 \times 29 = 232$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 14 • FORWARD 1 • Q2/5

431

QUESTION ID

S14-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 14 • FORWARD 1 • Q2/5

$6 \times 57 = ?$

A. 342

B. 63

C. 302

D. 352

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 14 • FORWARD 1 • Q3/5

QUESTION ID

S14-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 342

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 14 • FORWARD 1 • Q3/5

Multiply.

9 × 84

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 14 • FORWARD 1 • Q4/5

433

QUESTION ID

S14-F1-Q4

TYPE

Short answer

CORRECT ANSWER

756

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 14 • FORWARD 1 • Q4/5

True or false?

$4 \times 73 = 282$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 14 • FORWARD 1 • Q5/5

434

QUESTION ID

S14-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 14 • FORWARD 1 • Q5/5

SET 14 • 3.OA.C.7

FORWARD 2

Multiply multi-digit whole numbers with the standard algorithm • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5

Use the standard algorithm to multiply. Show your work.

346 × 27

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 14 • FORWARD 2 • Q1/5

436

QUESTION ID

S14-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

9,342

GRADING NOTE REQUIRED WORK: The standard multiplication algorithm with the partial products or regrouping shown. A correct product without the algorithm earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 14 • FORWARD 2 • Q1/5

Multiply.

58 × 64

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 14 • FORWARD 2 • Q2/5

437

QUESTION ID

S14-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3,712

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 14 • FORWARD 2 • Q2/5

214 × 35 = ?

A. 1,712

B. 7,390

C. 7,590

D. 7,490

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 14 • FORWARD 2 • Q3/5

438

QUESTION ID

S14-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7,490

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 14 • FORWARD 2 • Q3/5

True or false?

125 × 48 = 6,000

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 14 • FORWARD 2 • Q4/5

439

QUESTION ID

S14-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 14 • FORWARD 2 • Q4/5

True or false?

63 × 47 = 2,861

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 14 • FORWARD 2 • Q5/5

440

QUESTION ID

S14-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 14 • FORWARD 2 • Q5/5

SET 15 • 3.OA.C.7 • OPERATIONS & ALGEBRAIC THINKING

Divide fluently within 100

MAIN	Grade 3 main question	**3.OA.C.7**
BACKWARD 2	Split up to 20 objects into two equal groups	**2.OA.C.3**
BACKWARD 3	Equal rows in arrays up to 5 by 5	**2.OA.C.4**
FORWARD 1	Divide by a one-digit number, with remainders	**4.NBT.B.6**
FORWARD 2	Divide by a two-digit divisor	**5.NBT.B.6**

SET 15 • 3.OA.C.7

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Divide.

63 ÷ 7

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 15 • MAIN • Q1/5

443

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 15 • MAIN • Q1/5

Divide.

48 ÷ 8

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 15 • MAIN • Q2/5

444

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 15 • MAIN • Q2/5

54 ÷ 6 = ?

A. 48

B. 8

C. 9

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 15 • MAIN • Q3/5

445

QUESTION ID

S15-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 15 • MAIN • Q3/5

True or false?

72 ÷ 9 = 8

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 15 • MAIN • Q4/5

446

QUESTION ID

S15-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 15 • MAIN • Q4/5

True or false?

42 ÷ 6 = 8

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 15 • MAIN • Q5/5

447

QUESTION ID

S15-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 15 • MAIN • Q5/5

SET 15 • 3.OA.C.7

BACKWARD 2

Split up to 20 objects into two equal groups • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

16 counters are split into 2 equal groups.

How many counters are in each group?

Counters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 15 • BACKWARD 2 • Q1/5

449

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 15 • BACKWARD 2 • Q1/5

True or false?

20 objects make 2 equal groups of 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 15 • BACKWARD 2 • Q2/5

450

QUESTION ID

S15-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 15 • BACKWARD 2 • Q2/5

14 objects are split into 2 equal groups.

How many objects are in each group?

A. 28

B. 6

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 15 • BACKWARD 2 • Q3/5

451

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 15 • BACKWARD 2 • Q3/5

12 objects are split into 2 equal groups.

How many objects are in each group?

Objects: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 15 • BACKWARD 2 • Q4/5

452

QUESTION ID

S15-B2-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 15 • BACKWARD 2 • Q4/5

True or false?

18 objects make 2 equal groups of 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 15 • BACKWARD 2 • Q5/5

453

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • 3.OA.C.7

BACKWARD 3

Equal rows in arrays up to 5 by 5 • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

Write an addition equation with equal addends for the array.



Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 15 • BACKWARD 3 • Q1/5

455

QUESTION ID

S15-B3-Q1

TYPE

Short answer

CORRECT ANSWER

5 + 5 + 5 = 15

GRADING NOTE 3 + 3 + 3 + 3 + 3 = 15 (adding the columns) is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 15 • BACKWARD 3 • Q1/5

True or false?

5 + 5 + 5 + 5 = 20 matches an array of 4 rows with 5 dots in each row.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 15 • BACKWARD 3 • Q2/5

456

QUESTION ID

S15-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 15 • BACKWARD 3 • Q2/5

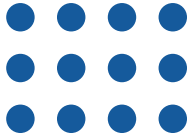
How many dots are in each row of the array?

A. 3

B. 4

C. 12

D. 7



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 15 • BACKWARD 3 • Q3/5

457

QUESTION ID

S15-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 15 • BACKWARD 3 • Q3/5

How many dots are in an array of 5 rows with 2 dots in each row?

Dots: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 15 • BACKWARD 3 • Q4/5

458

QUESTION ID

S15-B3-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 15 • BACKWARD 3 • Q4/5

True or false?

An array of 4 rows with 3 dots in each row has 7 dots.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 15 • BACKWARD 3 • Q5/5

459

QUESTION ID

S15-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B3-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • 3.OA.C.7

FORWARD 1

Divide by a one-digit number, with remainders • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Divide. Write the quotient and the remainder.

58 ÷ 7

Quotient: _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • FORWARD 1 • Q1/5

461

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8 R 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • FORWARD 1 • Q1/5

True or false?

76 ÷ 9 = 8 R 4

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • FORWARD 1 • Q2/5

462

QUESTION ID

S15-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • FORWARD 1 • Q2/5

Divide.

93 ÷ 4

A. 23

B. 23 R 1

C. 24 R 1

D. 22 R 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • FORWARD 1 • Q3/5

463

QUESTION ID

S15-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 23 R 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • FORWARD 1 • Q3/5

Divide.

87 ÷ 3

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • FORWARD 1 • Q4/5

464

QUESTION ID

S15-F1-Q4

TYPE

Short answer

CORRECT ANSWER

29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • FORWARD 1 • Q4/5

True or false?

65 ÷ 8 = 7 R 9

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • FORWARD 1 • Q5/5

465

QUESTION ID

S15-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 3.OA.C.7

FORWARD 2

Divide by a two-digit divisor • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

Divide.

864 ÷ 24

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • FORWARD 2 • Q1/5

467

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • FORWARD 2 • Q1/5

True or false?

1,104 ÷ 46 = 24

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • FORWARD 2 • Q2/5

468

QUESTION ID

S15-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • FORWARD 2 • Q2/5

Divide.

975 ÷ 39

A. 25

B. 26

C. 24

D. 936

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • FORWARD 2 • Q3/5

469

QUESTION ID

S15-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • FORWARD 2 • Q3/5

Divide. Write the quotient and the remainder.

700 ÷ 32

Quotient: _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • FORWARD 2 • Q4/5

470

QUESTION ID

S15-F2-Q4

TYPE

Short answer

CORRECT ANSWER

21 R 28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • FORWARD 2 • Q4/5

True or false?

1,512 ÷ 27 = 54

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • FORWARD 2 • Q5/5

471

QUESTION ID

S15-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • FORWARD 2 • Q5/5

SET 16 • 3.OA.C.7 • OPERATIONS & ALGEBRAIC THINKING

Use the relationship between multiplication and division

MAIN	Grade 3 main question	**3.OA.C.7**
BACKWARD 1	Use the relationship between addition and subtraction within 20	**1.OA.C.6**
BACKWARD 2	Subtract within 20 mentally	**2.OA.B.2**
FORWARD 1	Use multiplication to check division, including remainders	**4.NBT.B.6**
FORWARD 2	Divide a unit fraction by a whole number and check with multipli…	**5.NF.B.7.a**

SET 16 • 3.OA.C.7

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Write the four facts in the fact family for 6, 9, and 54.

Facts: _____

_____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • MAIN • Q1/5

474

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

6 × 9 = 54,
9 × 6 = 54,
54 ÷ 6 = 9,
54 ÷ 9 = 6

GRADING NOTE All four facts are required, in any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • MAIN • Q1/5

$7 \times 8 = 56$.

Which division fact is in the same fact family?

A. $56 \div 7 = 8$

B. $56 \times 7 = 8$

C. $7 \div 56 = 8$

D. $8 \div 56 = 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • MAIN • Q2/5

475

QUESTION ID

S16-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 56 ÷ 7 = 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • MAIN • Q2/5

True or false?

Because 4 × 9 = 36, you know that 36 ÷ 4 = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • MAIN • Q3/5

476

QUESTION ID

S16-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • MAIN • Q3/5

Use multiplication to check: is 63 ÷ 9 = 7 correct? Write the multiplication fact you used.

Correct? _____

Fact: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • MAIN • Q4/5

477

QUESTION ID

S16-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes; 9 × 7 = 63

GRADING NOTE 7 × 9 = 63 is also correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • MAIN • Q4/5

True or false?

Because 5 × 8 = 40, you know that 40 ÷ 8 = 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • MAIN • Q5/5

478

QUESTION ID

S16-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • MAIN • Q5/5

SET 16 • 3.OA.C.7

BACKWARD 1

Use the relationship between addition and subtraction within 20 • Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Write the four facts in the fact family for 7, 8, and 15.

Facts: _____

_____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 16 • BACKWARD 1 • Q1/5

480

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7 + 8 = 15,
8 + 7 = 15,
15 − 7 = 8,
15 − 8 = 7

GRADING NOTE All four facts are required, in any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 16 • BACKWARD 1 • Q1/5

True or false?

Because 9 + 4 = 13, you know that 13 − 4 = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 16 • BACKWARD 1 • Q2/5

481

QUESTION ID

S16-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 16 • BACKWARD 1 • Q2/5

6 + 5 = 11.

Which subtraction fact is in the same fact family?

A. 6 − 5 = 1

B. 11 − 5 = 6

C. 11 − 4 = 7

D. 5 − 6 = 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 16 • BACKWARD 1 • Q3/5

482

QUESTION ID

S16-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 11 − 5 = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 16 • BACKWARD 1 • Q3/5

Is 16 − 7 = 9 correct? Write the addition fact you used to check.

Correct? _____

Fact: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 16 • BACKWARD 1 • Q4/5

483

QUESTION ID

S16-B1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes; 7 + 9 = 16

GRADING NOTE 9 + 7 = 16 is also correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 16 • BACKWARD 1 • Q4/5

True or false?

Because 8 + 6 = 14, you know that 14 − 6 = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 16 • BACKWARD 1 • Q5/5

484

QUESTION ID

S16-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • 3.OA.C.7

BACKWARD 2

Subtract within 20 mentally • Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2

Subtract.

17 − 9

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 16 • BACKWARD 2 • Q1/5

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 16 • BACKWARD 2 • Q1/5

True or false?

$14 - 7 = 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 16 • BACKWARD 2 • Q2/5

487

QUESTION ID

S16-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 16 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

12 − 5 = ?

A. 7

B. 8

C. 17

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 16 • BACKWARD 2 • Q3/5

488

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 16 • BACKWARD 2 • Q3/5

Subtract.

13 − 8

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 16 • BACKWARD 2 • Q4/5

489

QUESTION ID

S16-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 16 • BACKWARD 2 • Q4/5

True or false?

16 − 8 = 9

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 16 • BACKWARD 2 • Q5/5

490

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.B.2 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 3.OA.C.7

FORWARD 1

Use multiplication to check division, including remainders • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Check: is 47 ÷ 5 = 9 R 2 correct? Write the multiplication and addition you used.

Correct? _____

Check: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 16 • FORWARD 1 • Q1/5

492

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes; 5 × 9 + 2 = 47

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 16 • FORWARD 1 • Q1/5

True or false?

34 ÷ 6 = 5 R 4 because 6 × 5 + 4 = 34.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 16 • FORWARD 1 • Q2/5

493

QUESTION ID

S16-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 16 • FORWARD 1 • Q2/5

Which equation shows that 75 ÷ 8 = 9 R 3?

A. 8 + 9 + 3 = 75

B. 8 × 9 = 75

C. 9 × 3 + 8 = 75

D. 8 × 9 + 3 = 75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 16 • FORWARD 1 • Q3/5

494

QUESTION ID

S16-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8 × 9 + 3 = 75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 16 • FORWARD 1 • Q3/5

Use multiplication to check: is 96 ÷ 4 = 24 correct?
Write the multiplication you used.

Correct? _____

Check: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 16 • FORWARD 1 • Q4/5

495

QUESTION ID

S16-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes; 4 × 24 = 96

GRADING NOTE 24 × 4 = 96 is also correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 16 • FORWARD 1 • Q4/5

True or false?

59 ÷ 7 = 8 R 2 because 7 × 8 + 2 = 59.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 16 • FORWARD 1 • Q5/5

496

QUESTION ID

S16-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 16 • FORWARD 1 • Q5/5

SET 16 • 3.OA.C.7

FORWARD 2

Divide a unit fraction by a whole number and check with multiplication • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a

Find $\frac{1}{3} \div 4$. Then write the multiplication that checks it.

Quotient: _____

Check: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a • Set 16 • FORWARD 2 • Q1/5

498

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1/12; 1/12 × 4 = 1/3

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a • Set 16 • FORWARD 2 • Q1/5

True or false?

$\frac{1}{2} \div 3 = \frac{1}{6}$ because $\frac{1}{6} \times 3 = \frac{1}{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a • Set 16 • FORWARD 2 • Q2/5

499

QUESTION ID

S16-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a • Set 16 • FORWARD 2 • Q2/5

Which equation checks $\frac{1}{4} \div 2 = \frac{1}{8}$?

A. $\frac{1}{4} \times 2 = \frac{1}{8}$

B. $\frac{1}{8} \times 2 = \frac{1}{4}$

C. $2 \times 4 = 8$

D. $\frac{1}{8} \times \frac{1}{4} = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a • Set 16 • FORWARD 2 • Q3/5

500

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/8 × 2 = 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a • Set 16 • FORWARD 2 • Q3/5

Divide.

$\frac{1}{5} \div 2$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a • Set 16 • FORWARD 2 • Q4/5

501

QUESTION ID

S16-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a • Set 16 • FORWARD 2 • Q4/5

True or false?

$\frac{1}{6} \div 2 = \frac{1}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a • Set 16 • FORWARD 2 • Q5/5

QUESTION ID

S16-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.7.a • Set 16 • FORWARD 2 • Q5/5

SET 17 • 3.OA.D.8 • OPERATIONS & ALGEBRAIC THINKING

Solve two-step word problems using the four operations

MAIN	Grade 3 main question	**3.OA.D.8**
BACKWARD 1	One-step take-from word problems within 100	**2.OA.A.1**
BACKWARD 2	Find the total of equal rows by repeated addition	**2.OA.C.4**
FORWARD 1	Solve multistep word problems with whole numbers	**4.OA.A.3**
FORWARD 2	Write expressions that record multistep calculations	**5.OA.A.2**

SET 17 • 3.OA.D.8

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

Mia has 4 bags with 6 marbles in each bag. She gives away 9 marbles.

How many marbles does she have left?

Marbles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 17 • MAIN • Q1/5

505

QUESTION ID

S17-M-Q1

TYPE

Short answer

CORRECT ANSWER

15 marbles

GRADING NOTE $4 \times 6 = 24$ and $24 - 9 = 15$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 17 • MAIN • Q1/5

A class has 27 students, and 3 more students join. The class is split into 5 equal teams.

How many students are on each team?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 17 • MAIN • Q2/5

506

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

6 students

GRADING NOTE 27 + 3 = 30 and 30 ÷ 5 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 17 • MAIN • Q2/5

Pens cost $3 each. Leo buys 7 pens and pays with a $50 bill.

How much change does he get?

A. $21

B. $53

C. $47

D. $29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 17 • MAIN • Q3/5

507

QUESTION ID

S17-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 17 • MAIN • Q3/5

True or false?

A farmer has 32 eggs. He packs them 8 to a carton and sells 2 cartons. He has 2 cartons left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 17 • MAIN • Q4/5

508

QUESTION ID

S17-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 17 • MAIN • Q4/5

Ana read 8 pages on Monday. On Tuesday she read 4 sections of 8 pages each.

How many pages did she read on the two days?

Pages: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 17 • MAIN • Q5/5

509

QUESTION ID

S17-M-Q5

TYPE

Short answer

CORRECT ANSWER

40 pages

GRADING NOTE 4 × 8 = 32 and 8 + 32 = 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 17 • MAIN • Q5/5

SET 17 • 3.OA.D.8

BACKWARD 1

One-step take-from word problems within 100 • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

There were 72 people on a train. 26 got off.

How many people are still on the train?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 17 • BACKWARD 1 • Q1/5

511

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 17 • BACKWARD 1 • Q1/5

True or false?

Kim had 90 cents and spent 45 cents. She has 45 cents left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 17 • BACKWARD 1 • Q2/5

512

QUESTION ID

S17-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 17 • BACKWARD 1 • Q2/5

A store had 60 hats and sold 34.

How many hats are left?

A. 36

B. 26

C. 34

D. 94

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 17 • BACKWARD 1 • Q3/5

513

QUESTION ID

S17-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 17 • BACKWARD 1 • Q3/5

A tree had 83 apples. 47 fell off.

How many apples are still on the tree?

Apples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 17 • BACKWARD 1 • Q4/5

514

QUESTION ID

S17-B1-Q4

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 17 • BACKWARD 1 • Q4/5

True or false?

A box had 55 nails. 18 were used, so 47 are left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 17 • BACKWARD 1 • Q5/5

515

QUESTION ID

S17-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 3.OA.D.8

BACKWARD 2

Find the total of equal rows by repeated addition • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

There are 4 rows of 5 chairs.

Use repeated addition to find the number of chairs.

Chairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 17 • BACKWARD 2 • Q1/5

517

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE 5 + 5 + 5 + 5 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 17 • BACKWARD 2 • Q1/5

True or false?

3 rows of 4 tiles is 4 + 4 + 4 = 12 tiles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 17 • BACKWARD 2 • Q2/5

518

QUESTION ID

S17-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 17 • BACKWARD 2 • Q2/5

There are 5 rows of 2 trees.

Which addition gives the total?

A. 2 + 2 + 2 + 2 + 2

B. 5 + 2

C. 2 + 5 + 2

D. 5 + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 17 • BACKWARD 2 • Q3/5

519

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2 + 2 + 2 + 2 + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 17 • BACKWARD 2 • Q3/5

There are 3 rows of 3 cups.

How many cups are there?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 17 • BACKWARD 2 • Q4/5

520

QUESTION ID

S17-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 17 • BACKWARD 2 • Q4/5

True or false?

2 rows of 4 desks is 4 + 2 = 6 desks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 17 • BACKWARD 2 • Q5/5

521

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 3.OA.D.8

FORWARD 1

Solve multistep word problems with whole numbers • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

A store gets 6 boxes of 48 cans. It sells 175 cans.

How many cans are left?

Cans: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 17 • FORWARD 1 • Q1/5

523

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

113 cans

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 17 • FORWARD 1 • Q1/5

True or false?

Tickets cost $12 for adults and $7 for children. 3 adults and 4 children pay $64 in all.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 17 • FORWARD 1 • Q2/5

524

QUESTION ID

S17-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 17 • FORWARD 1 • Q2/5

A school has 4 classes of 26 students and 3 classes of 24 students.

How many students are in these classes?

A. 72

B. 160

C. 176

D. 104

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 17 • FORWARD 1 • Q3/5

525

QUESTION ID

S17-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 176

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 17 • FORWARD 1 • Q3/5

A library has 1,250 books. It buys 8 boxes of 35 books.

How many books does it have now?

Books: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 17 • FORWARD 1 • Q4/5

526

QUESTION ID

S17-F1-Q4

TYPE

Short answer

CORRECT ANSWER

1,530 books

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 17 • FORWARD 1 • Q4/5

True or false?

Leo saves $15 a week for 6 weeks and then spends $38. He has $62 left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 17 • FORWARD 1 • Q5/5

527

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 17 • FORWARD 1 • Q5/5

SET 17 • 3.OA.D.8

FORWARD 2

Write expressions that record multistep calculations • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for: buy 4 packs of 12 pencils, then give away 9 pencils.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 17 • FORWARD 2 • Q1/5

529

QUESTION ID

S17-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 × 12 − 9

GRADING NOTE (4 × 12) − 9 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 17 • FORWARD 2 • Q1/5

27 students are joined by 3 more, and then they are split into 5 equal teams.

Which expression matches?

A. (27 + 3) ÷ 5

B. 27 + 3 ÷ 5

C. 27 + (3 ÷ 5)

D. 5 ÷ (27 + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 17 • FORWARD 2 • Q2/5

530

QUESTION ID

S17-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. (27 + 3) ÷ 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 17 • FORWARD 2 • Q2/5

True or false?

The expression $50 - 3 \times 7$ gives the change from $50 after buying 7 pens that cost $3 each.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 17 • FORWARD 2 • Q3/5

531

QUESTION ID

S17-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 17 • FORWARD 2 • Q3/5

Write an expression for: add 18 and 6, divide the sum by 4, and then multiply by 3.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 17 • FORWARD 2 • Q4/5

532

QUESTION ID

S17-F2-Q4

TYPE

Short answer

CORRECT ANSWER

(18 + 6) ÷ 4 × 3

GRADING NOTE ((18 + 6) ÷ 4) × 3 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 17 • FORWARD 2 • Q4/5

True or false?

"Read 8 pages, then read 4 more sections of 8 pages" is matched by the expression (8 + 4) × 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 17 • FORWARD 2 • Q5/5

533

QUESTION ID

S17-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 17 • FORWARD 2 • Q5/5

SET 18 • 3.OA.D.8 • OPERATIONS & ALGEBRAIC THINKING

Represent two-step word problems with an equation with a letter for the unknown

MAIN	Grade 3 main question	**3.OA.D.8**
BACKWARD 1	Write an equation with a symbol for the unknown in a one-step wo…	**2.OA.A.1**
BACKWARD 2	Subtract two-digit numbers within 100	**2.NBT.B.5**
FORWARD 1	Write and solve equations with a letter for multistep problems	**4.OA.A.3**
FORWARD 2	Write and interpret numerical expressions without evaluating th…	**5.OA.A.2**

SET 18 • 3.OA.D.8

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

A baker makes 5 trays of 8 rolls. She sells 23 rolls.

Write an equation with a letter for the number of rolls left. Then solve it.

Equation: _____

Rolls left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 18 • MAIN • Q1/5

536

QUESTION ID

S18-M-Q1

TYPE

Short answer

CORRECT ANSWER

r = 5 × 8 − 23;

r = 17

GRADING NOTE Any equivalent equation with a letter for the unknown is correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 18 • MAIN • Q1/5

Sam has $40. He buys 3 books that cost $9 each.

Which equation gives the money he has left, m?

A. m = 40 − 3 + 9

B. m = 3 × 9 − 40

C. m = 40 + 3 × 9

D. m = 40 − 3 × 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 18 • MAIN • Q2/5

537

QUESTION ID

S18-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. m = 40 − 3 × 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 18 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

There are 36 cupcakes. 4 are eaten, and the rest are shared equally by 8 friends.

Write an equation with a letter for each friend's share. Then solve it.

Equation: _____

Each friend: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 18 • MAIN • Q3/5

538

QUESTION ID

S18-M-Q3

TYPE

Short answer

CORRECT ANSWER

c = (36 − 4) ÷ 8;

c = 4

GRADING NOTE Any equivalent equation with a letter for the unknown is correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 18 • MAIN • Q3/5

True or false?

"Jo had 15 stickers and then got 6 packs of 5 stickers" can be written as s = 15 + 6 × 5, and s = 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 18 • MAIN • Q4/5

539

QUESTION ID

S18-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 18 • MAIN • Q4/5

True or false?

"7 boxes of 6 crayons, then 10 crayons are lost" can be written as c = 7 + 6 − 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 18 • MAIN • Q5/5

540

QUESTION ID

S18-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 18 • MAIN • Q5/5

SET 18 • 3.OA.D.8

BACKWARD 1

Write an equation with a symbol for the unknown in a one-step word problem • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Ty had some marbles. He got 25 more and now has 61.

Write an equation with a symbol for the unknown.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 18 • BACKWARD 1 • Q1/5

542

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

? + 25 = 61

GRADING NOTE Any equation with a symbol for the unknown (such as m + 25 = 61 or 61 − 25 = m) is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 18 • BACKWARD 1 • Q1/5

True or false?

"44 ducks were on a pond, and 19 flew away" can be written as 44 − 19 = d.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 18 • BACKWARD 1 • Q2/5

543

QUESTION ID

S18-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 18 • BACKWARD 1 • Q2/5

There are 28 red cars and some blue cars. There are 50 cars in all.

Which equation matches?

A. b − 50 = 28

B. 50 + 28 = b

C. 28 − b = 50

D. 28 + b = 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 18 • BACKWARD 1 • Q3/5

544

QUESTION ID

S18-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 28 + b = 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 18 • BACKWARD 1 • Q3/5

A pole is 70 cm long. Lu cuts off some, and 42 cm is left.

Write an equation with a symbol for the unknown.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 18 • BACKWARD 1 • Q4/5

545

QUESTION ID

S18-B1-Q4

TYPE

Short answer

CORRECT ANSWER

70 − ? = 42

GRADING NOTE 70 − c = 42 and 42 + c = 70 are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 18 • BACKWARD 1 • Q4/5

True or false?

"There were 36 books, and 12 more came" can be written as 36 − 12 = b.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 18 • BACKWARD 1 • Q5/5

546

QUESTION ID

S18-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 3.OA.D.8

BACKWARD 2

Subtract two-digit numbers within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Subtract.

70 − 34

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 2 • Q1/5

548

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 2 • Q1/5

True or false?

58 − 29 = 29

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 2 • Q2/5

549

QUESTION ID

S18-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 2 • Q2/5

84 − 47 = ?

A. 37

B. 131

C. 43

D. 47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 2 • Q3/5

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 37

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 2 • Q3/5

Subtract.

65 − 38

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 2 • Q4/5

551

QUESTION ID

S18-B2-Q4

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 2 • Q4/5

True or false?

$93 - 55 = 48$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 2 • Q5/5

552

QUESTION ID

S18-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 3.OA.D.8

FORWARD 1

Write and solve equations with a letter for multistep problems • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

A club buys 6 packs of 15 cups and uses 58 cups.

Write an equation with a letter for the cups left. Then solve it.

Equation: _____

Cups left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 18 • FORWARD 1 • Q1/5

554

QUESTION ID

S18-F1-Q1

TYPE

Short answer

CORRECT ANSWER

c = 6 × 15 − 58;

c = 32

GRADING NOTE Any equivalent equation with a letter is correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 18 • FORWARD 1 • Q1/5

A farmer has 120 eggs. She sells 7 cartons of 12 eggs.

Which equation gives the eggs left, e?

A. e = 120 − 7 × 12

B. e = 120 − 7 + 12

C. e = 7 × 12 − 120

D. e = 120 ÷ 7 × 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 18 • FORWARD 1 • Q2/5

555

QUESTION ID

S18-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. e = 120 − 7 × 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 18 • FORWARD 1 • Q2/5

True or false?

After a $16 gift, the rest of the cost of a $96 bike is paid in 8 equal parts. This can be written as $p = (96 - 16) \div 8$, and $p = 10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 18 • FORWARD 1 • Q3/5

556

QUESTION ID

S18-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 18 • FORWARD 1 • Q3/5

4 vans carry 12 students each, and 3 cars carry 5 students each.

Write an equation with a letter for the number of students. Then solve it.

Equation: _____

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 18 • FORWARD 1 • Q4/5

557

QUESTION ID

S18-F1-Q4

TYPE

Short answer

CORRECT ANSWER

s = 4 × 12 + 3 × 5;

s = 63

GRADING NOTE Any equivalent equation with a letter is correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 18 • FORWARD 1 • Q4/5

True or false?

"A hall has 250 seats, and 9 rows of 24 seats are filled" can be written as s = 250 − 9 + 24 for the empty seats.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 18 • FORWARD 1 • Q5/5

558

QUESTION ID

S18-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 18 • FORWARD 1 • Q5/5

SET 18 • 3.OA.D.8

FORWARD 2

Write and interpret numerical expressions without evaluating them • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for "3 times the difference of 40 and 12."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 18 • FORWARD 2 • Q1/5

560

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3 × (40 − 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 18 • FORWARD 2 • Q1/5

True or false?

(24 + 16) ÷ 8 is one-eighth of 24 + 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 18 • FORWARD 2 • Q2/5

561

QUESTION ID

S18-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 18 • FORWARD 2 • Q2/5

Without calculating, how does 5 × (92 − 17) compare with 92 − 17?

A. It is 5 times as large.

B. It is 5 less.

C. It is 5 more.

D. It is the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 18 • FORWARD 2 • Q3/5

562

QUESTION ID

S18-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. It is 5 times as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 18 • FORWARD 2 • Q3/5

Write an expression for "add 15 and 9, then divide by 6."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 18 • FORWARD 2 • Q4/5

563

QUESTION ID

S18-F2-Q4

TYPE

Short answer

CORRECT ANSWER

(15 + 9) ÷ 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 18 • FORWARD 2 • Q4/5

True or false?

7 × (18 + 3) is 7 more than 18 + 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 18 • FORWARD 2 • Q5/5

564

QUESTION ID

S18-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 18 • FORWARD 2 • Q5/5

SET 19 • 3.OA.D.8 • OPERATIONS & ALGEBRAIC THINKING

Check whether answers to two-step word problems are reasonable, using mental computation and rounding

MAIN	Grade 3 main question	**3.OA.D.8**
BACKWARD 1	Add or subtract 10 or 100 mentally	**2.NBT.B.8**
BACKWARD 2	Compare three-digit numbers	**2.NBT.A.4**
FORWARD 1	Use rounding to check answers to multistep word problems	**4.OA.A.3**
FORWARD 2	Round decimals to estimate sums, differences, and products	**5.NBT.A.4**

SET 19 • 3.OA.D.8

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

A store had 512 apples. It sold 189 in the morning and 207 in the afternoon. Lee says 116 apples are left.

Round each number to the nearest hundred to estimate. Is his answer reasonable?

Estimate: _____

Reasonable? _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 19 • MAIN • Q1/5

567

QUESTION ID

S19-M-Q1

TYPE

Short answer

CORRECT ANSWER

100; yes

GRADING NOTE 500 − 200 − 200 = 100, which is close to 116 (the exact answer is 116). Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 19 • MAIN • Q1/5

A class collected 238 cans on Monday and 315 cans on Tuesday. Then it gave away 146 cans. Ana says 707 cans are left.

Round each number to the nearest hundred to estimate. Is her answer reasonable?

Estimate: _____

Reasonable? _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 19 • MAIN • Q2/5

568

QUESTION ID

S19-M-Q2

TYPE

Short answer

CORRECT ANSWER

400; no

GRADING NOTE 200 + 300 − 100 = 400, which is far from 707 (the exact answer is 407). Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 19 • MAIN • Q2/5

Ben buys 4 packs of 9 stickers. Then he gets 38 more stickers.

Keep 4 × 9 as it is and round 38 to the nearest ten. Which is the estimate of his total?

A. 80

B. 120

C. 36

D. 76

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 19 • MAIN • Q3/5

569

QUESTION ID

S19-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 76

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 19 • MAIN • Q3/5

True or false?

A farmer picks 391 apples and 412 pears, then sells 298 pieces of fruit. Rounding each number to the nearest hundred, about 500 pieces of fruit are left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 19 • MAIN • Q4/5

570

QUESTION ID

S19-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 19 • MAIN • Q4/5

True or false?

Kim had 85 cents. She spent 38 cents and then found 21 cents. Rounding each amount to the nearest ten, she has about 50 cents.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 19 • MAIN • Q5/5

571

QUESTION ID

S19-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 19 • MAIN • Q5/5

SET 19 • 3.OA.D.8

BACKWARD 1

Add or subtract 10 or 100 mentally • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8

What is 100 more than 453?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 19 • BACKWARD 1 • Q1/5

573

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

553

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 19 • BACKWARD 1 • Q1/5

True or false?

10 less than 280 is 270.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 19 • BACKWARD 1 • Q2/5

574

QUESTION ID

S19-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 19 • BACKWARD 1 • Q2/5

What is 100 less than 615?

A. 614

B. 715

C. 515

D. 605

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 19 • BACKWARD 1 • Q3/5

575

QUESTION ID

S19-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 515

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 19 • BACKWARD 1 • Q3/5

What is 10 more than 395?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 19 • BACKWARD 1 • Q4/5

576

QUESTION ID

S19-B1-Q4

TYPE

Short answer

CORRECT ANSWER

405

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 19 • BACKWARD 1 • Q4/5

True or false?

100 more than 860 is 870.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 19 • BACKWARD 1 • Q5/5

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.8 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • 3.OA.D.8

BACKWARD 2

Compare three-digit numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Write >, =, or <.

482 ___ 428

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 19 • BACKWARD 2 • Q1/5

579

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

>

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 19 • BACKWARD 2 • Q1/5

True or false?

609 < 690

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 19 • BACKWARD 2 • Q2/5

580

QUESTION ID

S19-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 19 • BACKWARD 2 • Q2/5

Which number is greatest?

A. 713

B. 731

C. 371

D. 317

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 19 • BACKWARD 2 • Q3/5

581

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 731

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 19 • BACKWARD 2 • Q3/5

Write >, =, or <.

350 ___ 305

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 19 • BACKWARD 2 • Q4/5

582

QUESTION ID

S19-B2-Q4

TYPE

Short answer

CORRECT ANSWER

>

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 19 • BACKWARD 2 • Q4/5

True or false?

298 > 302

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 19 • BACKWARD 2 • Q5/5

583

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 3.OA.D.8

FORWARD 1

Use rounding to check answers to multistep word problems • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3

Ty buys 6 boxes of 48 pens and gives away 95 pens. He says he has 193 pens left.

Keep 6 unchanged. Round 48 and 95 to the nearest ten. Estimate, then tell whether his answer is reasonable.

Estimate: _____

Reasonable? _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 19 • FORWARD 1 • Q1/5

585

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

200; yes

GRADING NOTE 6 × 50 − 100 = 200, which is close to 193 (the exact answer is 193). Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 19 • FORWARD 1 • Q1/5

True or false?

A farm has 1,212 chickens. It buys 389 more and then sells 498. Rounding each number to the nearest hundred, about 1,100 chickens are left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 19 • FORWARD 1 • Q2/5

586

QUESTION ID

S19-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 19 • FORWARD 1 • Q2/5

A coach buys 4 boxes of 39 balls and then 52 more balls.

Keep 4 unchanged. Round 39 and 52 to the nearest ten. Which is the estimate of 4 × 39 + 52?

A. 160

B. 210

C. 190

D. 200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 19 • FORWARD 1 • Q3/5

587

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 210

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 19 • FORWARD 1 • Q3/5

A theater sold 6,814 tickets on Friday and 2,390 on Saturday. Then 1,975 tickets were refunded.

Round each number to the nearest thousand to estimate the tickets kept. Is an answer of 7,229 reasonable?

Estimate: _____

Reasonable? _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 19 • FORWARD 1 • Q4/5

588

QUESTION ID

S19-F1-Q4

TYPE

Short answer

CORRECT ANSWER

7,000; yes

GRADING NOTE 7,000 + 2,000 − 2,000 = 7,000, which is close to 7,229 (the exact answer). Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 19 • FORWARD 1 • Q4/5

True or false?

Leo says 8 boxes of 21 pens plus 37 more pens is 405 pens. Keeping 8 unchanged and rounding 21 and 37 to the nearest ten, the estimate shows that his answer is reasonable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 19 • FORWARD 1 • Q5/5

589

QUESTION ID

S19-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.3 • Set 19 • FORWARD 1 • Q5/5

SET 19 • 3.OA.D.8

FORWARD 2

Round decimals to estimate sums, differences, and products • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round each number to the nearest whole number. Then estimate.

3.8 + 6.4

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 2 • Q1/5

591

QUESTION ID

S19-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 2 • Q1/5

True or false?

Rounding each number to the nearest whole number, 9.7 − 2.2 is about 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 2 • Q2/5

592

QUESTION ID

S19-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 2 • Q2/5

Round each number to the nearest whole number.

Which is the estimate of 5.6 × 3.1?

A. 16

B. 15

C. 20

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 2 • Q3/5

593

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 2 • Q3/5

Round each number to the nearest whole number. Then estimate.

7.4 × 2.8

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 2 • Q4/5

594

QUESTION ID

S19-F2-Q4

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 2 • Q4/5

True or false?

Rounding each number to the nearest whole number, 12.4 + 3.7 is about 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 2 • Q5/5

595

QUESTION ID

S19-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • FORWARD 2 • Q5/5

SET 20 • 3.OA.D.9 • OPERATIONS & ALGEBRAIC THINKING

Identify arithmetic patterns in the addition and multiplication tables

MAIN	Grade 3 main question	**3.OA.D.9**
BACKWARD 1	Odd and even numbers up to 20	**2.OA.C.3**
BACKWARD 2	Skip-count by 5s, 10s, and 100s and see patterns	**2.NBT.A.2**
FORWARD 1	Generate a pattern from a rule and identify its features	**4.OA.C.5**
FORWARD 2	Generate two numerical patterns and compare their terms	**5.OA.B.3**

SET 20 • 3.OA.D.9

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

True or false?

Every product in the 4s row of the multiplication table is even.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 20 • MAIN • Q1/5

598

QUESTION ID

S20-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 20 • MAIN • Q1/5

The 5s row of the multiplication table is 5, 10, 15, 20, 25, ...

What pattern do you see in the ones digits?

Pattern: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 20 • MAIN • Q2/5

599

QUESTION ID

S20-M-Q2

TYPE

Short answer

CORRECT ANSWER

The ones digits alternate between 5 and 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 20 • MAIN • Q2/5

In the 9s row of the multiplication table (9, 18, 27, 36, ...), what happens to the digits from one product to the next?

A. The tens digit goes up by 1 and the ones digit goes down by 1.

B. Both digits go up by 1.

C. The digits stay the same.

D. Both digits go down by 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 20 • MAIN • Q3/5

600

QUESTION ID

S20-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The tens digit goes up by 1 and the ones digit goes down by 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 20 • MAIN • Q3/5

In the addition table, look at the sums 1 + 9, 2 + 8, 3 + 7, and 4 + 6.

What do you notice?

Pattern: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 20 • MAIN • Q4/5

601

QUESTION ID

S20-M-Q4

TYPE

Short answer

CORRECT ANSWER

Every sum is 10.

GRADING NOTE Any description saying all these sums are 10 (one addend goes up by 1 while the other goes down by 1) is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 20 • MAIN • Q4/5

True or false?

Every product in the 3s row of the multiplication table is odd.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 20 • MAIN • Q5/5

602

QUESTION ID

S20-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 20 • MAIN • Q5/5

SET 20 • 3.OA.D.9

BACKWARD 1

Odd and even numbers up to 20 • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

Is 17 odd or even?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 20 • BACKWARD 1 • Q1/5

604

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Odd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 20 • BACKWARD 1 • Q1/5

True or false?

14 is even.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 20 • BACKWARD 1 • Q2/5

605

QUESTION ID

S20-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 20 • BACKWARD 1 • Q2/5

Which number is odd?

A. 16

B. 19

C. 12

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 20 • BACKWARD 1 • Q3/5

606

QUESTION ID

S20-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 20 • BACKWARD 1 • Q3/5

Write 14 as the sum of two equal addends.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 20 • BACKWARD 1 • Q4/5

607

QUESTION ID

S20-B1-Q4

TYPE

Short answer

CORRECT ANSWER

7 + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 20 • BACKWARD 1 • Q4/5

True or false?

An odd number of objects can be put into two equal groups with none left over.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 20 • BACKWARD 1 • Q5/5

608

QUESTION ID

S20-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • 3.OA.D.9

BACKWARD 2

Skip-count by 5s, 10s, and 100s and see patterns • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2

Skip-count by 10s from 230.

230, 240, 250, ___, ___

Next two: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 20 • BACKWARD 2 • Q1/5

610

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

260, 270

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 20 • BACKWARD 2 • Q1/5

True or false?

When you skip-count by 5s from 5, every number ends in 5 or 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 20 • BACKWARD 2 • Q2/5

611

QUESTION ID

S20-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 20 • BACKWARD 2 • Q2/5

What comes next?

400, 500, 600, ___

A. 601

B. 610

C. 800

D. 700

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 20 • BACKWARD 2 • Q3/5

612

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 700

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 20 • BACKWARD 2 • Q3/5

Skip-count by 5s from 65.

65, 70, 75, ___, ___

Next two: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 20 • BACKWARD 2 • Q4/5

613

QUESTION ID

S20-B2-Q4

TYPE

Short answer

CORRECT ANSWER

80, 85

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 20 • BACKWARD 2 • Q4/5

True or false?

When you skip-count by 10s from 7, every number ends in 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 20 • BACKWARD 2 • Q5/5

614

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.2 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 3.OA.D.9

FORWARD 1

Generate a pattern from a rule and identify its features • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "add 6," starting at 3.

Write the first five terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 20 • FORWARD 1 • Q1/5

616

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3, 9, 15, 21, 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 20 • FORWARD 1 • Q1/5

True or false?

The rule "add 4," starting at 1, makes only odd numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 20 • FORWARD 1 • Q2/5

617

QUESTION ID

S20-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 20 • FORWARD 1 • Q2/5

The rule is "multiply by 2," starting at 5.

What is the 4th term?

A. 20

B. 80

C. 13

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 20 • FORWARD 1 • Q3/5

618

QUESTION ID

S20-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 20 • FORWARD 1 • Q3/5

The rule is "add 5," starting at 2: 2, 7, 12, 17, ...

What do you notice about the ones digits?

Pattern: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 20 • FORWARD 1 • Q4/5

619

QUESTION ID

S20-F1-Q4

TYPE

Short answer

CORRECT ANSWER

They alternate between 2 and 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 20 • FORWARD 1 • Q4/5

True or false?

The rule "add 3," starting at 4, makes only even numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 20 • FORWARD 1 • Q5/5

620

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 20 • FORWARD 1 • Q5/5

SET 20 • 3.OA.D.9

FORWARD 2

Generate two numerical patterns and compare their terms • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 6.

Write the first four terms of each pattern.

Pattern A: _____

Pattern B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 20 • FORWARD 2 • Q1/5

622

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0, 3, 6, 9; 0, 6, 12, 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 20 • FORWARD 2 • Q1/5

True or false?

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10. Each term of B is 2 times the matching term of A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 20 • FORWARD 2 • Q2/5

623

QUESTION ID

S20-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 20 • FORWARD 2 • Q2/5

Pattern A: 0, 4, 8, 12. Pattern B: 0, 12, 24, 36.

How does each B term compare with the matching A term?

A. It is the same.

B. It is 4 times as large.

C. It is 3 times as large.

D. It is 8 more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 20 • FORWARD 2 • Q3/5

624

QUESTION ID

S20-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. It is 3 times as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 20 • FORWARD 2 • Q3/5

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 8.

What is the B term when the A term is 10?

B term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 20 • FORWARD 2 • Q4/5

625

QUESTION ID

S20-F2-Q4

TYPE

Short answer

CORRECT ANSWER

40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 20 • FORWARD 2 • Q4/5

True or false?

Pattern A: 0, 2, 4, 6. Pattern B: 0, 6, 12, 18. Each B term is 4 more than the matching A term.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 20 • FORWARD 2 • Q5/5

626

QUESTION ID

S20-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 20 • FORWARD 2 • Q5/5

SET 21 • 3.OA.D.9 • OPERATIONS & ALGEBRAIC THINKING

Explain arithmetic patterns using properties of operations

MAIN	Grade 3 main question	**3.OA.D.9**
BACKWARD 1	Explain why a number is even or odd	**2.OA.C.3**
BACKWARD 2	Explain that addends can be added in any order	**1.OA.B.3**
FORWARD 1	Explain features of a number pattern	**4.OA.C.5**
FORWARD 2	Explain the relationship between two numerical patterns	**5.OA.B.3**

SET 21 • 3.OA.D.9

MAIN

Grade 3 main question • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

ANSWER KEY Reduced copy of the question page, with its answer

Explain why 4 times any whole number is always even.

Explanation: _____

_____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 21 • MAIN • Q1/5

629

QUESTION ID

S21-M-Q1

TYPE

Short answer

CORRECT ANSWER

4 times a number is 2 times (2 times the number), so it can always be split into two equal groups.

GRADING NOTE Any correct reason using equal groups, doubling, or 4 × n = 2 × (2 × n) is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 21 • MAIN • Q1/5

Why is every product in the 6s row of the multiplication table even?

A. The 6s row has 10 products.

B. 6 is greater than 5.

C. 6 = 2 × 3, so each product is 2 times a number.

D. Every number is even.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 21 • MAIN • Q2/5

630

QUESTION ID

S21-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 6 = 2 × 3, so each product is 2 times a number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 21 • MAIN • Q2/5

True or false?

The sum of two even numbers is always even.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 21 • MAIN • Q3/5

631

QUESTION ID

S21-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 21 • MAIN • Q3/5

Is the product of two odd numbers odd or even? Give an example.

Answer: _____

Example: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 21 • MAIN • Q4/5

632

QUESTION ID

S21-M-Q4

TYPE

Short answer

CORRECT ANSWER

Odd; 3 × 5 = 15

GRADING NOTE Any correct example of two odd factors with an odd product is correct. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 21 • MAIN • Q4/5

True or false?

An odd number times 2 is always odd.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 21 • MAIN • Q5/5

633

QUESTION ID

S21-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 21 • MAIN • Q5/5

SET 21 • 3.OA.D.9

BACKWARD 1

Explain why a number is even or odd • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3

Explain why 12 is even.

Explanation: _____

_____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 21 • BACKWARD 1 • Q1/5

635

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 = 6 + 6, so it is the sum of two equal addends (it splits into 2 equal groups).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 21 • BACKWARD 1 • Q1/5

True or false?

An even number can be written as the sum of two equal addends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 21 • BACKWARD 1 • Q2/5

636

QUESTION ID

S21-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 21 • BACKWARD 1 • Q2/5

Which equation shows that 10 is even?

A. 10 = 10 + 0

B. 10 = 7 + 3

C. 10 = 5 + 5

D. 10 = 9 + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 21 • BACKWARD 1 • Q3/5

637

QUESTION ID

S21-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10 = 5 + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 21 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 15 even or odd? Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 21 • BACKWARD 1 • Q4/5

638

QUESTION ID

S21-B1-Q4

TYPE

Short answer

CORRECT ANSWER

Odd

GRADING NOTE 15 cannot be split into two equal groups: in pairs, one is left over. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 21 • BACKWARD 1 • Q4/5

True or false?

7 can be written as the sum of two equal whole-number addends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 21 • BACKWARD 1 • Q5/5

639

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.3 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 3.OA.D.9

BACKWARD 2

Explain that addends can be added in any order • Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3

Explain why 6 + 9 = 9 + 6.

Explanation: _____

_____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 21 • BACKWARD 2 • Q1/5

641

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Changing the order of the addends does not change the sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 21 • BACKWARD 2 • Q1/5

True or false?

5 + 8 = 8 + 5

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 21 • BACKWARD 2 • Q2/5

642

QUESTION ID

S21-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 21 • BACKWARD 2 • Q2/5

Which equation shows adding in a different order?

A. 4 + 7 = 47

B. 4 + 7 = 7 + 4

C. 4 + 7 = 4 − 7

D. 4 + 7 = 7 − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 21 • BACKWARD 2 • Q3/5

643

QUESTION ID

S21-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4 + 7 = 7 + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 21 • BACKWARD 2 • Q3/5

9 + 3 = 12.

What is 3 + 9?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 21 • BACKWARD 2 • Q4/5

644

QUESTION ID

S21-B2-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 21 • BACKWARD 2 • Q4/5

True or false?

$8 + 2 = 2 + 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 21 • BACKWARD 2 • Q5/5

645

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.3 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 3.OA.D.9

FORWARD 1

Explain features of a number pattern • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "add 4," starting at 2: 2, 6, 10, 14, ...

Why is every term even?

Explanation: _____

_____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • FORWARD 1 • Q1/5

647

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

It starts at an even number, and adding an even number (4) to an even number always gives an even number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • FORWARD 1 • Q1/5

True or false?

The rule "add 5," starting at 1, makes terms that go odd, even, odd, even.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • FORWARD 1 • Q2/5

648

QUESTION ID

S21-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • FORWARD 1 • Q2/5

The rule is "add 10," starting at 3.

Why does every term end in 3?

A. Adding 10 does not change the ones digit.

B. The terms get larger.

C. 3 is odd.

D. 10 is even.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • FORWARD 1 • Q3/5

649

QUESTION ID

S21-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Adding 10 does not change the ones digit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The rule is "multiply by 3," starting at 1: 1, 3, 9, 27, ...

Are the terms odd or even? Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • FORWARD 1 • Q4/5

650

QUESTION ID

S21-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Odd

GRADING NOTE The pattern starts at an odd number, and an odd number times 3 (an odd number) is odd. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • FORWARD 1 • Q4/5

True or false?

The rule "add 2," starting at 5, makes only even numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • FORWARD 1 • Q5/5

651

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 3.OA.D.9

FORWARD 2

Explain the relationship between two numerical patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 6.

Explain why each B term is 3 times the matching A term.

Explanation: _____

_____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 21 • FORWARD 2 • Q1/5

653

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Both start at 0 and take the same number of steps, and B adds 3 times as much (6 = 3 × 2) at each step.

GRADING NOTE Must connect the rule amounts (6 = 3 × 2) with the same start of 0 and the same number of steps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 21 • FORWARD 2 • Q1/5

True or false?

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 15. Each B term is 3 times the matching A term.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 21 • FORWARD 2 • Q2/5

654

QUESTION ID

S21-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 21 • FORWARD 2 • Q2/5

Pattern A starts at 1 and adds 1. Pattern B starts at 1 and adds 2.

Why is B NOT always 2 times A?

A. A has more terms.

B. Both rules add.

C. B adds less than A.

D. They both start at 1, not at 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 21 • FORWARD 2 • Q3/5

655

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. They both start at 1, not at 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 21 • FORWARD 2 • Q3/5

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 12.

What is the A term when the B term is 36?

A term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 21 • FORWARD 2 • Q4/5

656

QUESTION ID

S21-F2-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 21 • FORWARD 2 • Q4/5

True or false?

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9. Each B term is 6 more than the matching A term.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 21 • FORWARD 2 • Q5/5

657

QUESTION ID

S21-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 21 • FORWARD 2 • Q5/5

SET 22 • 3.NBT.A.1 • NUMBER & OPERATIONS IN BASE TEN

Round whole numbers to the nearest 10

MAIN	Grade 3 main question	**3.NBT.A.1**
BACKWARD 1	Hundreds, tens, and ones in three-digit numbers	**2.NBT.A.1**
BACKWARD 2	Whole numbers on a number line	**2.MD.B.6**
FORWARD 1	Round multi-digit whole numbers to any place	**4.NBT.A.3**
FORWARD 2	Round decimals to any place	**5.NBT.A.4**

SET 22 • 3.NBT.A.1

MAIN

Grade 3 main question • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1

Round 47 to the nearest ten.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 22 • MAIN • Q1/5

660

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 22 • MAIN • Q1/5

True or false?

118 rounded to the nearest ten is 110.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 22 • MAIN • Q2/5

661

QUESTION ID

S22-M-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 22 • MAIN • Q2/5

Round 685 to the nearest ten.

A. 600

B. 680

C. 690

D. 700

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 22 • MAIN • Q3/5

662

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 690

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 22 • MAIN • Q3/5

True or false?

234 rounded to the nearest ten is 230.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 22 • MAIN • Q4/5

663

QUESTION ID

S22-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 22 • MAIN • Q4/5

Use the number line.

Is 76 closer to 70 or to 80? Round 76 to the nearest ten.



Closer to: _____

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 22 • MAIN • Q5/5

664

QUESTION ID

S22-M-Q5

TYPE

Short answer

CORRECT ANSWER

80; 80

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 22 • MAIN • Q5/5

SET 22 • 3.NBT.A.1

BACKWARD 1

Hundreds, tens, and ones in three-digit numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1

What digit is in the tens place of 563?

Digit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 22 • BACKWARD 1 • Q1/5

666

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 22 • BACKWARD 1 • Q1/5

True or false?

In 708, the 0 means 0 tens.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 22 • BACKWARD 1 • Q2/5

667

QUESTION ID

S22-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 22 • BACKWARD 1 • Q2/5

Which number has 3 hundreds, 8 tens, and 2 ones?

A. 328

B. 832

C. 3,082

D. 382

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 22 • BACKWARD 1 • Q3/5

668

QUESTION ID

S22-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 382

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 22 • BACKWARD 1 • Q3/5

What is the value of the 4 in 245?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 22 • BACKWARD 1 • Q4/5

669

QUESTION ID

S22-B1-Q4

TYPE

Short answer

CORRECT ANSWER

40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 22 • BACKWARD 1 • Q4/5

True or false?

In 961, the 6 is worth 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 22 • BACKWARD 1 • Q5/5

670

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.1 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 3.NBT.A.1

BACKWARD 2

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 22 • BACKWARD 2 • Q1/5

672

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 22 • BACKWARD 2 • Q1/5

True or false?

On a number line, 68 is between 60 and 70.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 22 • BACKWARD 2 • Q2/5

673

QUESTION ID

S22-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 22 • BACKWARD 2 • Q2/5

Which number is closest to 30 on a number line?

A. 32

B. 37

C. 39

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 22 • BACKWARD 2 • Q3/5

674

QUESTION ID

S22-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 22 • BACKWARD 2 • Q3/5

What number is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 22 • BACKWARD 2 • Q4/5

675

QUESTION ID

S22-B2-Q4

TYPE

Short answer

CORRECT ANSWER

87

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

On a number line, 54 is closer to 60 than to 50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 22 • BACKWARD 2 • Q5/5

676

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 3.NBT.A.1

FORWARD 1

Round multi-digit whole numbers to any place • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3

Round 3,467 to the nearest ten.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 22 • FORWARD 1 • Q1/5

678

QUESTION ID

S22-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3,470

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 22 • FORWARD 1 • Q1/5

True or false?

12,845 rounded to the nearest hundred is 12,800.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 22 • FORWARD 1 • Q2/5

679

QUESTION ID

S22-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 22 • FORWARD 1 • Q2/5

Round 58,349 to the nearest thousand.

A. 58,000

B. 60,000

C. 59,000

D. 58,300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 22 • FORWARD 1 • Q3/5

680

QUESTION ID

S22-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 58,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 22 • FORWARD 1 • Q3/5

Round 7,996 to the nearest ten.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 22 • FORWARD 1 • Q4/5

681

QUESTION ID

S22-F1-Q4

TYPE

Short answer

CORRECT ANSWER

8,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 22 • FORWARD 1 • Q4/5

True or false?

41,705 rounded to the nearest ten thousand is 50,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 22 • FORWARD 1 • Q5/5

682

QUESTION ID

S22-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 22 • FORWARD 1 • Q5/5

SET 22 • 3.NBT.A.1

FORWARD 2

Round decimals to any place • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 4.67 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 22 • FORWARD 2 • Q1/5

684

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 22 • FORWARD 2 • Q1/5

True or false?

2.349 rounded to the nearest hundredth is 2.35.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 22 • FORWARD 2 • Q2/5

685

QUESTION ID

S22-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 22 • FORWARD 2 • Q2/5

Round 8.52 to the nearest whole number.

A. 8.6

B. 9

C. 8

D. 8.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 22 • FORWARD 2 • Q3/5

686

QUESTION ID

S22-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 22 • FORWARD 2 • Q3/5

Round 0.938 to the nearest hundredth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 22 • FORWARD 2 • Q4/5

687

QUESTION ID

S22-F2-Q4

TYPE

Short answer

CORRECT ANSWER

0.94

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 22 • FORWARD 2 • Q4/5

True or false?

6.45 rounded to the nearest tenth is 6.4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 22 • FORWARD 2 • Q5/5

688

QUESTION ID

S22-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 22 • FORWARD 2 • Q5/5

SET 23 • 3.NBT.A.1 • NUMBER & OPERATIONS IN BASE TEN

Round whole numbers to the nearest 100

MAIN	Grade 3 main question	**3.NBT.A.1**
BACKWARD 1	Read and write numbers to 1000 in expanded form	**2.NBT.A.3**
BACKWARD 2	Three-digit numbers on a number line	**2.MD.B.6**
FORWARD 1	Round multi-digit whole numbers to the hundreds, thousands, an…	**4.NBT.A.3**
FORWARD 2	Round decimals to the nearest whole number, tenth, or hundredth	**5.NBT.A.4**

SET 23 • 3.NBT.A.1

MAIN

Grade 3 main question • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1

Round 382 to the nearest hundred.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 23 • MAIN • Q1/5

691

QUESTION ID

S23-M-Q1

TYPE

Short answer

CORRECT ANSWER

400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 23 • MAIN • Q1/5

True or false?

749 rounded to the nearest hundred is 700.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 23 • MAIN • Q2/5

692

QUESTION ID

S23-M-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 23 • MAIN • Q2/5

Round 650 to the nearest hundred.

A. 700

B. 610

C. 650

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 23 • MAIN • Q3/5

693

QUESTION ID

S23-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 700

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 23 • MAIN • Q3/5

True or false?

216 rounded to the nearest hundred is 300.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 23 • MAIN • Q4/5

694

QUESTION ID

S23-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 23 • MAIN • Q4/5

A school has 538 students.

Round the number of students to the nearest hundred.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 23 • MAIN • Q5/5

695

QUESTION ID

S23-M-Q5

TYPE

Short answer

CORRECT ANSWER

500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.1 • Set 23 • MAIN • Q5/5

SET 23 • 3.NBT.A.1

BACKWARD 1

Read and write numbers to 1000 in expanded form • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3

Write 627 in expanded form.

Expanded form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 23 • BACKWARD 1 • Q1/5

697

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

600 + 20 + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 23 • BACKWARD 1 • Q1/5

True or false?

Four hundred five is written 405.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 23 • BACKWARD 1 • Q2/5

698

QUESTION ID

S23-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 23 • BACKWARD 1 • Q2/5

Which number is 300 + 70 + 4?

A. 734

B. 374

C. 3,074

D. 347

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 23 • BACKWARD 1 • Q3/5

699

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 374

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 23 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the number eight hundred sixty with digits.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 23 • BACKWARD 1 • Q4/5

700

QUESTION ID

S23-B1-Q4

TYPE

Short answer

CORRECT ANSWER

860

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 23 • BACKWARD 1 • Q4/5

True or false?

500 + 9 = 590

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 23 • BACKWARD 1 • Q5/5

701

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

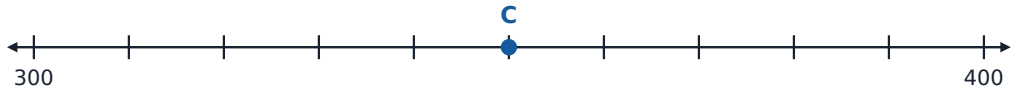
Grade 2 • Number & Operations in Base Ten • 2.NBT.A.3 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 3.NBT.A.1

BACKWARD 2

Three-digit numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point C?



C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 2 • Q1/5

703

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

350

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 2 • Q1/5

True or false?

On a number line, 720 is between 700 and 800.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 2 • Q2/5

704

QUESTION ID

S23-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 2 • Q2/5

Which number is closer to 600 than to 500?

A. 510

B. 520

C. 540

D. 580

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 2 • Q3/5

705

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 580

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 2 • Q3/5

What number is halfway between 200 and 300 on a number line?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 2 • Q4/5

706

QUESTION ID

S23-B2-Q4

TYPE

Short answer

CORRECT ANSWER

250

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 2 • Q4/5

True or false?

On a number line, 430 is closer to 500 than to 400.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 2 • Q5/5

707

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 3.NBT.A.1

FORWARD 1

Round multi-digit whole numbers to the hundreds, thousands, and beyond • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3

Round 6,251 to the nearest hundred.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 23 • FORWARD 1 • Q1/5

709

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6,300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 23 • FORWARD 1 • Q1/5

True or false?

94,512 rounded to the nearest thousand is 95,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 23 • FORWARD 1 • Q2/5

710

QUESTION ID

S23-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 23 • FORWARD 1 • Q2/5

Round 238,670 to the nearest hundred thousand.

A. 240,000

B. 300,000

C. 200,000

D. 238,700

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 23 • FORWARD 1 • Q3/5

711

QUESTION ID

S23-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 200,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 23 • FORWARD 1 • Q3/5

Round 9,950 to the nearest hundred.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 23 • FORWARD 1 • Q4/5

712

QUESTION ID

S23-F1-Q4

TYPE

Short answer

CORRECT ANSWER

10,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 23 • FORWARD 1 • Q4/5

True or false?

3,049 rounded to the nearest hundred is 3,100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 23 • FORWARD 1 • Q5/5

713

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 23 • FORWARD 1 • Q5/5

SET 23 • 3.NBT.A.1

FORWARD 2

Round decimals to the nearest whole number, tenth, or hundredth • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Round 13.48 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 23 • FORWARD 2 • Q1/5

715

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 23 • FORWARD 2 • Q1/5

True or false?

0.75 rounded to the nearest tenth is 0.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 23 • FORWARD 2 • Q2/5

716

QUESTION ID

S23-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 23 • FORWARD 2 • Q2/5

Round 2.963 to the nearest tenth.

A. 2.8

B. 2.96

C. 2.9

D. 3.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 23 • FORWARD 2 • Q3/5

717

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 23 • FORWARD 2 • Q3/5

Round 5.096 to the nearest hundredth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 23 • FORWARD 2 • Q4/5

718

QUESTION ID

S23-F2-Q4

TYPE

Short answer

CORRECT ANSWER

5.10

GRADING NOTE 5.1 names the same value and is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 23 • FORWARD 2 • Q4/5

True or false?

9.04 rounded to the nearest tenth is 9.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 23 • FORWARD 2 • Q5/5

719

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 23 • FORWARD 2 • Q5/5

SET 24 • 3.NBT.A.2 • NUMBER & OPERATIONS IN BASE TEN

Add within 1000 using place value strategies and algorithms

MAIN	Grade 3 main question	**3.NBT.A.2**
BACKWARD 1	Add within 100	**2.NBT.B.5**
BACKWARD 2	Add within 1000 by adding hundreds, tens, and ones	**2.NBT.B.7**
FORWARD 1	Add multi-digit whole numbers with the standard algorithm	**4.NBT.B.4**
FORWARD 2	Add decimals to hundredths	**5.NBT.B.7**

SET 24 • 3.NBT.A.2

MAIN

Grade 3 main question • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Add. Show your work using place value or the standard algorithm.

358 + 276

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 24 • MAIN • Q1/5

722

QUESTION ID

S24-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

634

GRADING NOTE REQUIRED WORK: Any valid place-value strategy or the standard algorithm, with the regrouping shown (for example 300 + 200 = 500, 50 + 70 = 120, 8 + 6 = 14, and 500 + 120 + 14 = 634). The sum alone earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 24 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Add.

407 + 395

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 24 • MAIN • Q2/5

723

QUESTION ID

S24-M-Q2

TYPE

Short answer

CORRECT ANSWER

802

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 24 • MAIN • Q2/5

529 + 263 = ?

A. 782

B. 792

C. 266

D. 702

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 24 • MAIN • Q3/5

724

QUESTION ID

S24-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 792

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 24 • MAIN • Q3/5

True or false?

186 + 614 = 800

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 24 • MAIN • Q4/5

QUESTION ID

S24-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 24 • MAIN • Q4/5

True or false?

475 + 348 = 813

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 24 • MAIN • Q5/5

726

QUESTION ID

S24-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 24 • MAIN • Q5/5

SET 24 • 3.NBT.A.2

BACKWARD 1

Add within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

58 + 36

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 24 • BACKWARD 1 • Q1/5

728

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

94

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 24 • BACKWARD 1 • Q1/5

True or false?

47 + 25 = 72

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 24 • BACKWARD 1 • Q2/5

729

QUESTION ID

S24-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 24 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

29 + 64 = ?

A. 94

B. 83

C. 93

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 24 • BACKWARD 1 • Q3/5

730

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 93

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 24 • BACKWARD 1 • Q3/5

Add.

75 + 18

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 24 • BACKWARD 1 • Q4/5

731

QUESTION ID

S24-B1-Q4

TYPE

Short answer

CORRECT ANSWER

93

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 24 • BACKWARD 1 • Q4/5

True or false?

66 + 27 = 83

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 24 • BACKWARD 1 • Q5/5

732

QUESTION ID

S24-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 3.NBT.A.2

BACKWARD 2

Add within 1000 by adding hundreds, tens, and ones • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7

Add using hundreds, tens, and ones.

234 + 152

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 24 • BACKWARD 2 • Q1/5

734

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

386

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 24 • BACKWARD 2 • Q1/5

True or false?

300 + 400 = 700

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 24 • BACKWARD 2 • Q2/5

735

QUESTION ID

S24-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 24 • BACKWARD 2 • Q2/5

Which shows the sum 245 + 132 by place value?

A. 200 + 70 + 7

B. 300 + 70 + 7

C. 300 + 70 + 5

D. 300 + 60 + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 24 • BACKWARD 2 • Q3/5

736

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 300 + 70 + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 24 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Add.

420 + 160

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 24 • BACKWARD 2 • Q4/5

737

QUESTION ID

S24-B2-Q4

TYPE

Short answer

CORRECT ANSWER

580

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 24 • BACKWARD 2 • Q4/5

True or false?

513 + 241 = 744

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 24 • BACKWARD 2 • Q5/5

738

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 3.NBT.A.2

FORWARD 1

Add multi-digit whole numbers with the standard algorithm • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Use the standard algorithm to add. Show your work.

3,768 + 2,457

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 24 • FORWARD 1 • Q1/5

740

QUESTION ID

S24-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

6,225

GRADING NOTE REQUIRED WORK: The standard addition algorithm with the regrouping shown. A correct sum without the algorithm earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 24 • FORWARD 1 • Q1/5

Add.

45,092 + 18,659

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 24 • FORWARD 1 • Q2/5

741

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

63,751

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 24 • FORWARD 1 • Q2/5

2,806 + 4,395 = ?

A. 7,201

B. 7,191

C. 7,101

D. 6,201

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 24 • FORWARD 1 • Q3/5

742

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7,201

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 24 • FORWARD 1 • Q3/5

True or false?

9,999 + 1 = 10,000

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 24 • FORWARD 1 • Q4/5

743

QUESTION ID

S24-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 24 • FORWARD 1 • Q4/5

True or false?

6,485 + 3,729 = 10,114

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 24 • FORWARD 1 • Q5/5

744

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 24 • FORWARD 1 • Q5/5

SET 24 • 3.NBT.A.2

FORWARD 2

Add decimals to hundredths • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Add.

3.45 + 2.8

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 2 • Q1/5

746

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

6.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 2 • Q1/5

True or false?

$0.6 + 0.47 = 1.07$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 2 • Q2/5

747

QUESTION ID

S24-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 2 • Q2/5

Add.

12.5 + 3.75

A. 16.25

B. 15.80

C. 16.20

D. 4.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 2 • Q3/5

748

QUESTION ID

S24-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 16.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 2 • Q3/5

Add.

4.09 + 5.91

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 2 • Q4/5

749

QUESTION ID

S24-F2-Q4

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE 10.00 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 2 • Q4/5

True or false?

$2.7 + 1.35 = 3.12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 2 • Q5/5

750

QUESTION ID

S24-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 24 • FORWARD 2 • Q5/5

SET 25 • 3.NBT.A.2 • NUMBER & OPERATIONS IN BASE TEN

Subtract within 1000 using place value strategies and algorithms

MAIN	Grade 3 main question	**3.NBT.A.2**
BACKWARD 1	Subtract within 100	**2.NBT.B.5**
BACKWARD 2	Subtract within 1000 by subtracting hundreds, tens, and ones	**2.NBT.B.7**
FORWARD 1	Subtract multi-digit whole numbers with the standard algorithm	**4.NBT.B.4**
FORWARD 2	Subtract decimals to hundredths	**5.NBT.B.7**

SET 25 • 3.NBT.A.2

MAIN

Grade 3 main question • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Subtract. Show your work using place value or the standard algorithm.

623 − 358

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 25 • MAIN • Q1/5

753

QUESTION ID

S25-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

265

GRADING NOTE REQUIRED WORK: Any valid place-value strategy or the standard algorithm, with the regrouping shown (for example counting up: 358 + 2 = 360, 360 + 40 = 400, 400 + 223 = 623, so 2 + 40 + 223 = 265). The difference alone earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 25 • MAIN • Q1/5

Subtract.

700 − 246

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 25 • MAIN • Q2/5

754

QUESTION ID

S25-M-Q2

TYPE

Short answer

CORRECT ANSWER

454

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 25 • MAIN • Q2/5

$815 - 379 = ?$

A. 546

B. 1,194

C. 436

D. 444

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 25 • MAIN • Q3/5

755

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 436

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 25 • MAIN • Q3/5

True or false?

$902 - 518 = 384$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 25 • MAIN • Q4/5

QUESTION ID

S25-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 25 • MAIN • Q4/5

True or false?

560 − 287 = 383

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 25 • MAIN • Q5/5

757

QUESTION ID

S25-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 25 • MAIN • Q5/5

SET 25 • 3.NBT.A.2

BACKWARD 1

Subtract within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Subtract.

82 − 45

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 25 • BACKWARD 1 • Q1/5

759

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

37

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 25 • BACKWARD 1 • Q1/5

True or false?

$60 - 23 = 37$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 25 • BACKWARD 1 • Q2/5

760

QUESTION ID

S25-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 25 • BACKWARD 1 • Q2/5

74 − 39 = ?

A. 35

B. 25

C. 113

D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 25 • BACKWARD 1 • Q3/5

761

QUESTION ID

S25-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 25 • BACKWARD 1 • Q3/5

Subtract.

53 − 17

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 25 • BACKWARD 1 • Q4/5

762

QUESTION ID

S25-B1-Q4

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 25 • BACKWARD 1 • Q4/5

True or false?

$91 - 58 = 43$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 25 • BACKWARD 1 • Q5/5

763

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • 3.NBT.A.2

BACKWARD 2

Subtract within 1000 by subtracting hundreds, tens, and ones • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7

Subtract using hundreds, tens, and ones.

587 − 243

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 25 • BACKWARD 2 • Q1/5

765

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

344

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 25 • BACKWARD 2 • Q1/5

True or false?

800 − 300 = 500

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 25 • BACKWARD 2 • Q2/5

766

QUESTION ID

S25-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 25 • BACKWARD 2 • Q2/5

465 − 120 = ?

A. 245

B. 355

C. 345

D. 585

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 25 • BACKWARD 2 • Q3/5

767

QUESTION ID

S25-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 345

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 25 • BACKWARD 2 • Q3/5

Subtract.

690 − 250

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 25 • BACKWARD 2 • Q4/5

768

QUESTION ID

S25-B2-Q4

TYPE

Short answer

CORRECT ANSWER

440

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 25 • BACKWARD 2 • Q4/5

True or false?

$758 - 426 = 342$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 25 • BACKWARD 2 • Q5/5

769

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.7 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 3.NBT.A.2

FORWARD 1

Subtract multi-digit whole numbers with the standard algorithm • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Use the standard algorithm to subtract. Show your work.

7,203 − 4,568

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 25 • FORWARD 1 • Q1/5

771

QUESTION ID

S25-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

2,635

GRADING NOTE REQUIRED WORK: The standard subtraction algorithm with the regrouping shown. A correct difference without the algorithm earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 25 • FORWARD 1 • Q1/5

Subtract.

50,000 − 23,418

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 25 • FORWARD 1 • Q2/5

772

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

26,582

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 25 • FORWARD 1 • Q2/5

8,412 − 3,756 = ?

A. 4,656

B. 5,344

C. 4,666

D. 4,756

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 25 • FORWARD 1 • Q3/5

773

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4,656

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 25 • FORWARD 1 • Q3/5

True or false?

$6{,}001 - 2{,}999 = 3{,}002$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 25 • FORWARD 1 • Q4/5

774

QUESTION ID

S25-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 25 • FORWARD 1 • Q4/5

True or false?

$9{,}350 - 4{,}875 = 4{,}575$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 25 • FORWARD 1 • Q5/5

775

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 25 • FORWARD 1 • Q5/5

SET 25 • 3.NBT.A.2

FORWARD 2

Subtract decimals to hundredths • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Subtract.

6.4 − 2.75

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 25 • FORWARD 2 • Q1/5

777

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3.65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 25 • FORWARD 2 • Q1/5

True or false?

$5 - 1.25 = 3.75$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 25 • FORWARD 2 • Q2/5

778

QUESTION ID

S25-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 25 • FORWARD 2 • Q2/5

Subtract.

$10.3 - 4.68$

A. 5.72

B. 14.98

C. 6.38

D. 5.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 25 • FORWARD 2 • Q3/5

779

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 25 • FORWARD 2 • Q3/5

Subtract.

8.06 − 3.9

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 25 • FORWARD 2 • Q4/5

780

QUESTION ID

S25-F2-Q4

TYPE

Short answer

CORRECT ANSWER

4.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 25 • FORWARD 2 • Q4/5

True or false?

$7.5 - 2.25 = 5.35$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 25 • FORWARD 2 • Q5/5

781

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 25 • FORWARD 2 • Q5/5

SET 26 • 3.NBT.A.2 • NUMBER & OPERATIONS IN BASE TEN

Use the relationship between addition and subtraction to subtract and to check answers

MAIN	Grade 3 main question	**3.NBT.A.2**
BACKWARD 1	Use addition facts to subtract within 20	**1.OA.C.6**
BACKWARD 2	Use the relationship between addition and subtraction within 100	**2.NBT.B.5**
FORWARD 1	Check multi-digit subtraction with addition	**4.NBT.B.4**
FORWARD 2	Check decimal subtraction with addition	**5.NBT.B.7**

SET 26 • 3.NBT.A.2

MAIN

Grade 3 main question • Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2

Find 500 − 287 by counting up from 287.

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 26 • MAIN • Q1/5

784

QUESTION ID

S26-M-Q1

TYPE

Short answer

CORRECT ANSWER

213

GRADING NOTE 287 + 3 = 290, 290 + 10 = 300, 300 + 200 = 500; 3 + 10 + 200 = 213.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 26 • MAIN • Q1/5

Which addition checks 642 − 275 = 367?

A. 275 + 642 = 367

B. 367 + 275 = 642

C. 367 + 642 = 275

D. 642 + 275 = 367

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 26 • MAIN • Q2/5

785

QUESTION ID

S26-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 367 + 275 = 642

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 26 • MAIN • Q2/5

True or false?

Because 318 + 457 = 775, you know that 775 − 457 = 318.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 26 • MAIN • Q3/5

786

QUESTION ID

S26-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 26 • MAIN • Q3/5

Use addition to check: is 803 − 426 = 387 correct?
Show the addition you used.

Correct? _____

Check: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 26 • MAIN • Q4/5

787

QUESTION ID
S26-M-Q4

TYPE
Short answer

CORRECT ANSWER
No; 387 + 426 = 813

GRADING NOTE 387 + 426 = 813, not 803, so the answer is wrong (the correct difference is 377). Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 26 • MAIN • Q4/5

True or false?

Because 245 + 380 = 625, you know that 625 − 245 = 385.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 26 • MAIN • Q5/5

788

QUESTION ID

S26-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.2 • Set 26 • MAIN • Q5/5

SET 26 • 3.NBT.A.2

BACKWARD 1

Use addition facts to subtract within 20 • Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6

Find 16 − 9 by thinking 9 + ? = 16.

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 26 • BACKWARD 1 • Q1/5

790

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 26 • BACKWARD 1 • Q1/5

True or false?

11 − 4 = 7 because 4 + 7 = 11.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 26 • BACKWARD 1 • Q2/5

791

QUESTION ID

S26-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 26 • BACKWARD 1 • Q2/5

Which addition fact checks 15 − 6 = 9?

A. 9 + 6 = 15

B. 9 − 6 = 3

C. 6 + 6 = 12

D. 15 + 6 = 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 26 • BACKWARD 1 • Q3/5

792

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9 + 6 = 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 26 • BACKWARD 1 • Q3/5

Write the addition fact that checks 18 − 9 = 9.

Fact: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 26 • BACKWARD 1 • Q4/5

793

QUESTION ID

S26-B1-Q4

TYPE

Short answer

CORRECT ANSWER

9 + 9 = 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 26 • BACKWARD 1 • Q4/5

True or false?

$12 - 8 = 5$ because $8 + 5 = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 26 • BACKWARD 1 • Q5/5

794

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.C.6 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 3.NBT.A.2

BACKWARD 2

Use the relationship between addition and subtraction within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Find 70 − 45 by counting up from 45.

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 26 • BACKWARD 2 • Q1/5

796

QUESTION ID

S26-B2-Q1

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 26 • BACKWARD 2 • Q1/5

True or false?

Because 36 + 28 = 64, you know that 64 − 28 = 36.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 26 • BACKWARD 2 • Q2/5

797

QUESTION ID

S26-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 26 • BACKWARD 2 • Q2/5

Which addition checks 90 − 37 = 53?

A. 90 + 37 = 53

B. 53 + 90 = 37

C. 37 + 90 = 53

D. 53 + 37 = 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 26 • BACKWARD 2 • Q3/5

798

QUESTION ID

S26-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 53 + 37 = 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 26 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 81 − 34 = 47 correct? Write the addition you used to check.

Correct? _____

Check: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 26 • BACKWARD 2 • Q4/5

799

QUESTION ID

S26-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Yes; 47 + 34 = 81

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 26 • BACKWARD 2 • Q4/5

True or false?

Because 42 + 19 = 61, you know that 61 − 42 = 29.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 26 • BACKWARD 2 • Q5/5

800

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 26 • BACKWARD 2 • Q5/5